The Price of Readout: Exact Interface Floors and Frozen-Readout Shift Evaluation

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Abstract

Models judged by held-out test performance can hide failure modes that appear only when the data distribution shifts after deployment. This paper traces one such blind spot to the *readout interface*, the restricted family of functions used to extract answers from a representation, and prices it exactly. We prove that a representation can retain every bit of task information while a restricted readout remains strictly uncertain, with exact floors (0.69 bits for XOR under linear-threshold readouts; 0.54 bits for a sparse-superposition family), and we determine in closed form what beating a floor must cost: for the superposition family, exactly the budgets with an unshared stored-parameter budget of at least 5 and at least 8 arithmetic operations. Pre-registered experiments (constructed Gaussian channels, finite-atom readout ladders, and CIFAR-10 CNNs trained with a controlled spurious color cue) test the predictions. The registered pass/fail criteria are met; one descriptive prediction fails and usefully exposes cue cannibalization. Cue-reliant models beat a cue-decorrelated control in standard testing at every coupling strength, then lost up to 3.6 bits under a cue-breaking shift while the control moved at most 0.02 bits, and the damage separates measurably into an interface part and a representation part. We recommend reporting frozen-versus-refit evaluation pairs: testing alone cannot price the readout. The two failure modes the pair separates also appear on a pretrained language model (Pythia-410m): low-bit weight quantization is information-borne, with an irreducible refit floor and a sharp four-to-three-bit onset, while distribution shift is interface-borne, recoverable in proportion to how far out-of-distribution the shift is.

1 Introduction

A machine-learning system compresses twice. A representation map turns raw inputs into features, discarding detail; a *readout* (a linear probe, a shallow decoder, a thresholded score) turns features into answers, and can only use what its restricted functional form lets it reach. That the information *usable* by a restricted observer differs from the information *present* is prior art: it is the subject of predictive $\mathcal{V}$-information [14] and the premise of the probing literature (Section 6.2). What this paper adds is not a theory of usable information; it prices the observer-dependence exactly, on families small enough to close, and audits it under distribution shift. The paper is a small exact-theory and measurement-protocol paper, and its claim has six steps:

1. *The observer-dependence of usable information is prior art, owned as such.* Restricted-family information quantities originate with Xu et al. [14]; probe capacity as a confound of representation content is the probing literature's starting point (Section 6.2). We price that observer-dependence exactly and audit it under shift; we do not rederive it.
2. *This paper specializes to hard readout partitions and declared K/E accounting.* The readout-constrained task uncertainty $H_{\mathcal{A}}(Y \mid Z) = \inf_{r\in\mathcal{A}} H(Y \mid r(Z))$ (Definition 2.1) is the calibrated-partition specialization of the usable-information idea, paired with counted stored parameters K and operations E under conventions declared before use (Section 2.4), the specialization chosen so that floors and budgets become exactly computable.
3. *Exact floors exist even under perfect sufficiency.* A representation can retain every bit of task information ($H(Y \mid Z) = 0$) while a restricted readout remains strictly uncertain by an exactly computed amount: $\frac{3}{4}h_2(\frac{1}{3}) \approx 0.6887$ bits for XOR under linear-threshold readouts (Theorem 4.4); $\frac{2}{3}h_2(\frac{1}{4}) \approx 0.5409$ bits for a one-dimensional sparse-superposition instance, unimproved by width-2 threshold networks and zero at width 3 (Theorem 4.5).
4. *The cost of beating a floor can be closed exactly in small families.* Within affine-threshold com-

positions, the XOR floor first falls at exactly $E = 12$ operations (Proposition 4.14); the superposition floor at exactly $E = 8$ operations and $K_{\text{unsh}} = 5$ unshared stored reals, and that family's entire payment frontier is the quadrant $\mathcal{F} = \{(K_{\text{unsh}}, E) : K_{\text{unsh}} \geq 5,\ E \geq 8\}$ (Propositions 4.19 and 4.20). Universal versions of "beating a floor must cost structure" are false: a quadratic decoder inside the declared operation basis and a two-parameter sine decoder outside it beat floors at essentially linear cost (Propositions 4.8 and 4.10). What remains open is stated as such (Open Problem 4.22).

5. *Frozen readouts under shift expose failures that refit evaluations hide.* A readout fitted under a distribution P and evaluated *frozen* under a shifted Q measures a deployment property, $\mathrm{CE}_Q(r_P)$, that no entropy bounds and no in-distribution measurement constrains. In a constructed Gaussian channel, an insufficiency of analytic size 0.0004 bits (invisible to any realistic test) cost ≈ 1.7 bits of frozen-readout transfer cross-entropy when the nuisance–signal coupling flipped sign (Section 5.1). CIFAR-10 CNNs trained with a constructed class-indexed color cue beat a cue-decorrelated control in-distribution *at every coupling strength*, then lost up to 3.56 bits under a decorrelating shift while the control moved at most 0.021 bits (Section 5.3). This *masking* is hidden both by in-distribution testing and by any evaluation that silently refits the readout under the shifted distribution.
6. *Therefore: report frozen/refit evaluation pairs.* The frozen number is what deployment will experience; the refit number is what the representation still supports; their difference is the price of the readout interface, and no protocol that reports only one of the two can see it (Section 6).

Behind step 3 sits a classical dichotomy: compression preserves task-conditional entropy if and only if the compression map is sufficient, with the gap given by an explicit divergence identity (Section 3). The paper states the dichotomy for its accounting consequences and claims no novelty for it; the proofs are relocated to Appendix D, and a reader who takes that section as textbook material is reading it correctly.

Four quantities. Keeping the six steps well-posed requires four quantities that are routinely conflated when a representation is evaluated by "how well a probe does": the task uncertainty the representation itself leaves, $H(Y \mid Z)$; the best any readout in a class can do, $H_{\mathcal{A}}(Y \mid Z)$; the best cross-entropy any *soft* readout in the class attains under the training distribution, $\mathrm{CE}^{\star}_{\mathcal{A}}(P)$, which, not an entropy, is what in-distribution probing actually estimates; and the cross-entropy the readout fitted under P actually delivers under a shifted Q, $\mathrm{CE}_Q(r_P)$, a property of the deployed interface, bounded by nothing, and the quantity that collapses in the masking experiments. This four-quantity separation (Figure 1; Section 2 gives the definitions and exact relations) is the paper's operational framework, not a preliminary.

Roadmap. Section 2 fixes definitions, Sections 3 and 4 carry the theory (steps 3 and 4), Section 5 the pre-registered measurements (step 5; the measurement discipline is stated there once, in full), and Section 6 the reporting recommendation (step 6).

2 Setup and Definitions

2.1 Spaces, random variables, and standing assumptions

Let $(\Omega, \mathcal{F}, P)$ be a probability space carrying an input X with values in a standard Borel space $(\mathcal{X}, \mathcal{B}_{\mathcal{X}})$, a task target Y with values in a finite alphabet $\mathcal{Y}$, and a measurable map $\phi : \mathcal{X} \to \mathcal{Z}$ into a standard Borel space, with $Z = \phi(X)$. Standing assumptions:

(A1) $|\mathcal{Y}| < \infty$, so $H(Y) \leq \log_2 |\mathcal{Y}| < \infty$ and all conditional entropies below are finite. (Everything extends verbatim to countable $\mathcal{Y}$ with $H(Y) < \infty$; for general alphabets the mutual-information statements survive with the general definition of I [12], while the conditional-entropy phrasing requires differential-entropy caveats and is not asserted.)

(A2) No side channel: Z is produced from X alone. For deterministic ϕ this is automatic; for a stochastic representation (a Markov kernel from $\mathcal{X}$ to $\mathcal{Z}$) it is the structural requirement that $Y \to X \to Z$ is a Markov chain. Without (A2) the results are false and uninteresting ($Z = Y$ is not a representation of X).

Since $\mathcal{X}$ and $\mathcal{Z}$ are standard Borel, regular conditional distributions exist [8]; fix versions $p(y \mid x) = P(Y = y \mid X = x)$ and $p(y \mid z) = P(Y = y \mid Z = z)$, defined P_X- and P_Z-almost everywhere. Logarithms are base 2 (bits), $0 \log 0 = 0$, and $h_2(p) = -p \log_2 p - (1 - p) \log_2(1 - p)$ is binary entropy.

2.2 Four evaluation quantities

The four quantities of the introduction, now precisely. The first two are distribution-level entropies; the third is a property of a *fitted* interface and is not an

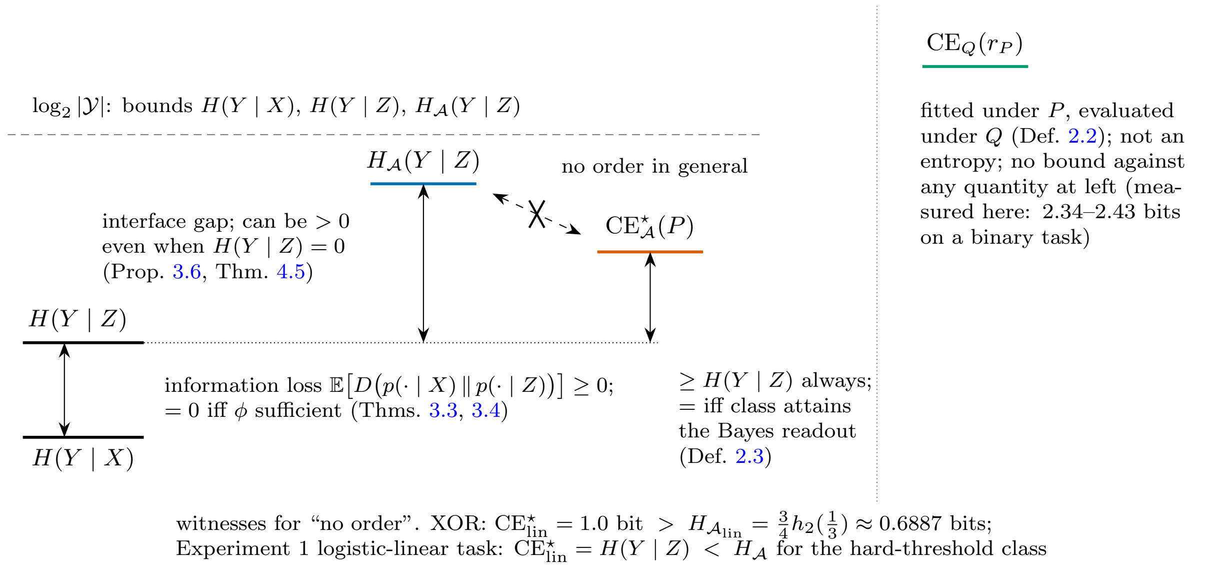


Figure 1: The four evaluation quantities of Section 2.2 and their exact relations (vertical position = bits, schematic, not to scale). Double-headed vertical arrows are the two proved gaps and the soft-optimum excess; the struck dashed arrow records that $H_{\mathcal{A}}(Y \mid Z)$ and $\mathrm{CE}^\star_{\mathcal{A}}(P)$ admit *no* order in general, with both witnesses shown (1.0 vs. 0.6887 bits for XOR, and the reverse strict order on the Experiment 1 logistic-linear construction). $\mathrm{CE}_Q(r_P)$, right of the dotted separator, is the only quantity evaluated under a shifted distribution Q; it obeys no entropy bound and in Section 5.1 exceeds the 1-bit binary bound by construction-predicted amounts.

entropy at all; the fourth is a population optimum over a soft readout class, the object the in-distribution experiments estimate. Conflating any two of them produces wrong conclusions: Section 5.1 measures three of the four (all but a hard-class $H_{\mathcal{A}}$), and Remark 3.7 separates the first two by theorem.

(1) Task-conditional entropy. Task-relevant uncertainty after representation is Shannon conditional entropy for the task's own target:

$$H(Y \mid Z) = \mathbb{E}\big[\mathcal{H}(p(\cdot \mid Z))\big],$$
$$\mathcal{H}(\mu) = -\textstyle\sum_{y \in \mathcal{Y}} \mu(y) \log \mu(y).$$

This quantity is a property of the pair (Y, Z) alone: no interface, no training. It satisfies $H(Y \mid Z) \leq \log_2|\mathcal{Y}|$ and is governed by the sufficiency dichotomy of Section 3.

(2) Readout-constrained task uncertainty.

Definition 2.1 (readout-constrained task uncertainty)**.** For a representation Z and a class $\mathcal{A}$ of measurable maps on $\mathcal{Z}$,

$$H_{\mathcal{A}}(Y \mid Z) := \inf_{r \in \mathcal{A}} H\big(Y \mid r(Z)\big).$$

Since each $r(Z)$ is a function of Z, the data-processing inequality gives $H_{\mathcal{A}}(Y \mid Z) \geq H(Y \mid Z)$ always, with equality whenever $\mathcal{A}$ contains a readout sufficient for Y given Z (e.g. the identity). Conditioning is on the σ-algebra $\sigma(r(Z))$; in particular a binary readout and its complement induce the same partition, so $H(Y \mid r(Z)) = H(Y \mid 1 - r(Z))$. The output cardinality of the class is part of its declaration; the parametric classes in this paper have binary outputs. $H_{\mathcal{A}}$ is still a population infimum (the best the class can do, not what any particular training run finds) and still satisfies $H_{\mathcal{A}}(Y \mid Z) \leq \log_2|\mathcal{Y}|$ (the constant readout is available in every class considered here). It is governed by the interface-floor results of Section 4.

(3) Frozen-readout transfer cross-entropy.

Definition 2.2 (frozen-readout transfer cross-entropy)**.** Let r_P be a probabilistic readout (a measurable map $z \mapsto q_{r_P}(\cdot \mid z)$, a distribution over $\mathcal{Y}$) selected (trained, calibrated, or population-optimized) using the distribution P, and let Q be an evaluation distribution for (Z, Y). The *transfer cross-entropy* of the frozen readout is

$$\mathrm{CE}_Q(r_P) := \mathbb{E}_Q\big[-\log q_{r_P}(Y \mid Z)\big].$$

Three properties make this a different object from (1) and (2). First, it depends on the fitted interface r_P, hence on the fitting distribution P, not only on the pair (Y, Z) under Q. Second, it obeys no entropy bound: $\mathrm{CE}_Q(r_P)$ can exceed $\log_2|\mathcal{Y}|$ arbitrarily (and in Section 5.1 it does: post-shift values of 2.34–2.43 bits for a binary task; in Section 5.3, 4.77 bits against the $\log_2 10 \approx 3.32$-bit ten-class bound). Third, it does not lower-bound or upper-bound $H_{\mathcal{A}}(Y \mid Z)$ under Q in general: a readout refit under Q (its population optimum is $\mathrm{CE}^{\star}_{\mathcal{A},Q}$, Definition 2.3 applied under Q) can sit far below $\mathrm{CE}_Q(r_P)$. The gap $\mathrm{CE}_Q(r_P) - \mathrm{CE}^{\star}_{\mathcal{A},Q}$ is *transfer risk* of the frozen interface, a deployment property, not an information property.

(4) Population soft-readout optimum.

Definition 2.3 (population soft-readout cross-entropy optimum)**.** For a class $\mathcal{A}_{\mathrm{soft}}$ of probabilistic readouts $q : z \mapsto q(\cdot \mid z)$ (each a distribution over $\mathcal{Y}$), the *population soft-readout optimum* under P is

$$\mathrm{CE}^{\star}_{\mathcal{A}}(P) := \inf_{q \in \mathcal{A}_{\mathrm{soft}}} \mathbb{E}_P\big[-\log q(Y \mid Z)\big].$$

$\mathrm{CE}^{\star}_{\mathcal{A}}$ is, up to notation and convention, the conditional $\mathcal{V}$-entropy of Xu et al. [14] for $\mathcal{V} = \mathcal{A}_{\mathrm{soft}}$; see Section 6.2.

The relations to the first two quantities are exact and asymmetric. For every probabilistic readout q,

$$\begin{aligned}\mathbb{E}_P\big[-\log q(Y \mid Z)\big] &= H(Y \mid Z) \\ &\quad + \mathbb{E}_P\big[D\big(p(\cdot \mid Z)\,\big\|\,q(\cdot \mid Z)\big)\big] \;\geq\; H(Y \mid Z),\end{aligned}$$

so $\mathrm{CE}^{\star}_{\mathcal{A}} \geq H(Y \mid Z)$ always, with equality when the class contains (or approaches) the Bayes readout $p(\cdot \mid z)$. But $\mathrm{CE}^{\star}_{\mathcal{A}}$ does *not* in general bound $H_{\mathcal{A}'}(Y \mid Z)$ for a hard deterministic class $\mathcal{A}'$, in either direction: a soft score is not a hard partition plus per-cell calibration. Both witnesses are recorded in Figure 1: for the XOR task of Proposition 3.6, $\mathrm{CE}^{\star}_{\mathrm{lin}} = 1$ bit exactly (the objective is convex and stationary at $w = 0$, $b = 0$), strictly *above* the hard floor $\frac{3}{4}h_2(\frac{1}{3}) \approx 0.6887$ bits; when $Y \mid Z$ is logistic-linear (the Experiment 1 construction), $\mathrm{CE}^{\star}_{\mathrm{lin}} = H(Y \mid Z)$, strictly *below* the hard-threshold class's $H_{\mathcal{A}}$. The experiments never assume the two objects coincide.

Measured quantities, and the bridge. The experiments of Section 5 measure held-out cross-entropies of trained soft readouts, written with the symbol CE throughout: $\widehat{\mathrm{CE}}_{\mathrm{ID}}$ for the in-distribution held-out cross-entropy of a readout trained and evaluated under P, and $\widehat{\mathrm{CE}}_Q(r_P)$ for the measured transfer cross-entropy under shift. The symbols H and $H_{\mathcal{A}}$ are reserved for the theorem-side quantities; $\mathrm{CE}^{\star}_{\mathcal{A}}$ (Definition 2.3) is what the in-distribution measurements estimate. The bridge is the identity displayed above: the *population* cross-entropy risk of a probabilistic readout upper-bounds $H(Y \mid Z)$, but never, in general, a hard-class $H_{\mathcal{A}}$; the measured held-out $\widehat{\mathrm{CE}}$ is a finite-sample estimate of that risk and may fluctuate below it. We treat $\widehat{\mathrm{CE}}$ as an estimate of its declared target $\mathrm{CE}^{\star}_{\mathcal{A}}$ (and, where the soft class provably contains the Bayes readout, of $H(Y \mid Z)$ itself) only where the estimator has been validated against analytic ground truth, which the in-distribution validation steps of Section 5 establish, to within the predicted maximum-likelihood estimation bias. No such validation is possible post-shift, which is why post-shift numbers are reported as $\mathrm{CE}_Q(r_P)$ and never as entropies. Figure 1 draws the four quantities with exactly these relations.

2.3 Lawfulness as sufficiency

Definition 2.4 (lawful compression)**.** ϕ is *lawful* (sufficient) for Y if $p(y \mid x) = p(y \mid \phi(x))$ for P_X-a.e. x and every $y \in \mathcal{Y}$. Writing $A := \{x \in \mathcal{X} : p(\cdot \mid x) \neq p(\cdot \mid \phi(x))\}$, ϕ is *non-sufficient* iff $P_X(A) > 0$.

A discrepancy confined to a P_X-null set is invisible to every quantity in this paper and counts as sufficiency; this null-set convention is exactly what makes Theorems 3.3 and 3.4 an exhaustive dichotomy.

2.4 Accounting conventions for K and E

Throughout, K denotes counted stored structure (parameter counts, codebook entries) and E counted operations (FLOPs per evaluation or per training run), each under a convention stated before use and, in the experiments, committed before any run. These are declared bookkeeping, not mathematics: different conventions give different constants, and we will be explicit about which statements are convention-robust and which are not.

Two stored-parameter conventions. For the circuit families of Section 4, two stored-parameter counts arise and must not be conflated. The *unshared count* K_{unsh} counts every gate coefficient and bias separately, whether or not stored values coincide numerically; the *distinct-value count* K_{val} counts distinct stored values once, however often a value is reused. Where a claim depends on the convention we write K_{unsh} or K_{val} explicitly; a bare K appears only where the convention in force has just been declared (as in each experiment's committed accounting) or where the two counts coincide.

Scope. The composite objective $\mathcal{R} = K + H_{\mathcal{T}} + \lambda E$ appears only in Corollary 3.8, where λ enters as a declared conversion rate between accounting units, quantified over and never assigned a value.

3 Lawful Compression: the Exact Dichotomy

Nothing in this section is new mathematics, and the paper does not rest on it for novelty: it is a clean restatement of the standard sufficiency/DPI equality conditions [1, 5, 12], organized (null-set discipline, explicit divergence form of the gap, and the accounting corollary) for the cost-accounting question the rest of the paper asks. The exact statements matter because the experiments of Section 5 are checked against them and the precision (a.s. conventions, which quantity is governed by which theorem) is load-bearing there; the proofs, which are classical, are relocated to Appendix D.

Lemma 3.1 (sufficiency $\Leftrightarrow$ conditional independence)**.** *ϕ is sufficient for Y (Definition 2.4) if and only if $Y \perp X \mid Z$, i.e. $P(Y = y \mid X, Z) = P(Y = y \mid Z)$ a.s. for every y.*

Lemma 3.2 (divergence identity)**.** *Under (A1)–(A2), with $Z = \phi(X)$,*

$$H(Y \mid Z) - H(Y \mid X) = \mathbb{E}\Big[D\big(p(\cdot \mid X) \,\big\|\, p(\cdot \mid Z)\big)\Big] \in [0, \log_2 |\mathcal{Y}|], \quad (1)$$

where $D(\mu\|\nu) = \sum_y \mu(y) \log \frac{\mu(y)}{\nu(y)}$ is relative entropy.

Identity (1) is the equality-condition form of the data-processing inequality specialized to this setup: the right side equals $I(Y; X \mid Z)$, and the chain rule $I(Y;X) = I(Y;Z) + I(Y;X \mid Z)$ together with $I(Y;Z \mid X) = 0$ (which is (A2)) recovers the DPI $I(Y;Z) \le I(Y;X)$ [1, Thm. 2.8.1]. Both theorems below fall out of (1) with no further work.

Theorem 3.3 (lawful compression is free)**.** *Assume (A1)–(A2) and let $Z = \phi(X)$. The data-processing inequality always gives $H(Y \mid Z) \ge H(Y \mid X)$ and $I(Y;Z) \le I(Y;X)$. If ϕ is sufficient for Y, both hold with equality:*

$$H(Y \mid Z) = H(Y \mid X), \qquad I(Y;Z) = I(Y;X).$$

Theorem 3.4 (unlawful compression strictly costs)**.** *Assume (A1)–(A2) and let $Z = \phi(X)$. If ϕ is not sufficient for Y, that is, $P_X(A) > 0$ for $A = \{x : p(\cdot \mid x) \ne p(\cdot \mid \phi(x))\}$, then*

$$H(Y \mid Z) > H(Y \mid X), \qquad I(Y;Z) < I(Y;X).$$

Remark 3.5 (null-set precision and exhaustiveness)*.* The dichotomy is exhaustive and exclusive because of the a.s. convention in Definition 2.4: discrepancy on a P_X-null set means sufficiency holds with a modified version and Theorem 3.3 applies; discrepancy on a positive-probability set means Theorem 3.4 applies. There is no intermediate case, and combining the two with Lemma 3.1 gives the standard DPI equality condition: $H(Y \mid Z) = H(Y \mid X)$ iff $Y \perp X \mid Z$.

Both theorems extend verbatim to stochastic representation maps (Markov kernels), with sufficiency generalized to $Y \perp X \mid Z$ (Remark D.1, Appendix D).

3.1 Sufficiency does not imply achievability: the XOR floor

Sufficiency says nothing about what a restricted readout class can extract. The following proposition makes that gap exact, with the classical XOR example [6] pushed through to the entropy floor; the proof (a finite enumeration, tabulated in Appendix A.1) is in Appendix D.

Proposition 3.6 (XOR floor under linear-threshold readouts)**.** *Let $X = (X_1, X_2)$ be uniform on $\{0,1\}^2$, $Y = X_1 \oplus X_2$, and $Z = X$ (the identity, trivially sufficient, so $H(Y \mid Z) = 0$). Let*

$$\mathcal{A}_{\text{lin}} = \big\{ z \mapsto \mathbf{1}\{w_1 z_1 + w_2 z_2 \ge \theta\} \;:\; w \in \mathbb{R}^2,\ \theta \in \mathbb{R} \big\}.$$

Then

$$H_{\mathcal{A}_{\text{lin}}}(Y \mid Z) = \tfrac{3}{4}\, h_2\big(\tfrac{1}{3}\big) = \tfrac{3}{4}\log_2 3 - \tfrac{1}{2} \approx 0.6887 \text{ bits} > 0 = H(Y \mid Z),$$

and in error-probability terms: Bayes error from Z is 0, while the best error achievable through any $\mathcal{A}_{\text{lin}}$-readout is $1/4$.

Remark 3.7 (lawful compression can strictly degrade the interface)*.* Theorem 3.3 does *not* survive the substitution $H \mapsto H_{\mathcal{A}}$. Let $X' = (X_1, X_2, X_1 \oplus X_2)$ with (X_1, X_2) uniform on $\{0,1\}^2$, $Y = X_1 \oplus X_2$, and $\phi(X') = (X_1, X_2)$ (drop the redundant third coordinate). ϕ is sufficient (the third coordinate is a function of the first two, so $p(y \mid X') = p(y \mid \phi(X'))$ pointwise), yet with $\mathcal{A}_{\text{lin}}$ the class of linear-threshold readouts on the available coordinates,

$$H_{\mathcal{A}_{\text{lin}}}(Y \mid X') = 0,$$
$$H_{\mathcal{A}_{\text{lin}}}(Y \mid \phi(X')) = \tfrac{3}{4} h_2(\tfrac{1}{3}) > 0,$$

the first by reading the third coordinate ($w = (0,0,1)$, $\theta = \frac{1}{2}$), the second by Proposition 3.6. Lawful compression preserved $H(Y \mid \cdot)$ exactly while strictly degrading what the restricted interface can extract. Lawfulness in the sense of Definition 2.4 is a statement about information, not about interface-accessible information; the two coincide only when $\mathcal{A}$ is rich enough (e.g. contains the Bayes readout for both representations). This distinction is a design requirement for measurement: an experiment must declare what its cross-entropy proxy targets: $H(Y \mid Z)$, a hard-class $H_{\mathcal{A}}(Y \mid Z)$, or a soft-class optimum $\mathrm{CE}^{\star}_{\mathcal{A}}$ (Definition 2.3). The three are governed by different statements. All three experiments in Section 5 declare it (their declared target is $\mathrm{CE}^{\star}_{\mathcal{A}}$ for the stated soft class; in Experiment 3 only *differences* of $\mathrm{CE}^{\star}$ are validated, Section 5.3).

3.2 Accounting consequence

Corollary 3.8 (K and E may drop freely under lawfulness)**.** *Let M_X and M_Z be representational processes operating on X and on $Z = \phi(X)$, with declared costs $(K(M_X), E(M_X))$ and $(K(M_Z), E(M_Z))$, and let the task term be $H_{\mathcal{T}} = H(Y \mid \cdot)$. If ϕ is sufficient for Y, then the objective $\mathcal{R} = K + H_{\mathcal{T}} + \lambda E$ changes by*

$$\Delta\mathcal{R} = \Delta K + \underbrace{\Delta H_{\mathcal{T}}}_{=0\ (\textit{Thm. } 3.3)} + \lambda\,\Delta E,$$

so any strict reduction in K and/or E (the other not increasing) strictly reduces $\mathcal{R}$ for every $\lambda \geq 0$, at exactly zero task-relevant-uncertainty cost. Conversely, if ϕ is non-sufficient, every such substitution pays $\Delta H_{\mathcal{T}} > 0$ (Theorem 3.4), and whether $\mathcal{R}$ decreases becomes a quantitative trade-off rather than a free move.

This is bookkeeping given the theorems (it contains no new mathematics), but it is the exact form in which the experiments of Section 5 check lawful-compression claims, and the "for every $\lambda \geq 0$" quantifier is what allows the check to avoid assigning λ any value.

4 Readout Geometry: Exact Interface Floors and the Price of Beating Them

This section carries the paper's theorem-side content. Here Z is fixed and the readout class varies. The first statement is trivial and is labeled as such; the content is the exact strict gaps, the payment analysis within the hidden-layer threshold ladder, the two counterexamples that refute the universal forms of the accounting claim (Section 4.4), and the payment frontier, closed exactly for the superposition family (Section 4.6).

Proposition 4.1 (monotonicity under class inclusion)**.** *If $\mathcal{A}_1 \subseteq \mathcal{A}_2$, then $H_{\mathcal{A}_2}(Y \mid Z) \leq H_{\mathcal{A}_1}(Y \mid Z)$.*

Proof. The infimum over a superset of a set of reals is no larger. □

The inequality can hold with equality under *strict* inclusion (Theorem 4.5(iii) proves an instance), so "larger readout classes need not help" is itself theorem-backed.

4.1 Declared parametric classes and their accounting

The payment analysis needs declared, countable K (stored parameters) and E (inference FLOPs per evaluation) per class. These are stated conventions: a length-d dot product costs $2d-1$ FLOPs, subtracting a threshold costs 1, a comparison costs 1. Every K value in this section and in the ladder declarations below is an *unshared* count, $K = K_{\mathrm{unsh}}$ in the Section 2.4 notation; where value sharing changes a claim we mark the shared count K_{val} explicitly.

$$\mathcal{A}_{\mathrm{lin}}(d): \quad z \mapsto \mathbf{1}\{w \cdot z \geq \theta\}$$
$$K = d+1, \quad E = 2d+1$$

$$\mathcal{A}^{\mathrm{thr}}_{m}(d): \quad z \mapsto \mathbf{1}\{\textstyle\sum_{i=1}^{m} v_i\, g_i(z) + c \geq 0\}$$
$$K = m(d{+}1) + (m{+}1), \quad E = m(2d{+}1) + (2m{+}1)$$

where each hidden unit is itself a linear threshold, $g_i(z) = \mathbf{1}\{w_i \cdot z \geq \theta_i\}$. We call the family $\{\mathcal{A}^{\mathrm{thr}}_m(d)\}_m$ the *hidden-layer threshold ladder*: the output unit reads the hidden units *only*: no unit reads a raw coordinate of z alongside hidden-unit outputs. This is the family for which the payment floors of Proposition 4.7 are proved; it must not be confused with the strictly more permissive *affine-threshold compositions* $\mathcal{A}_{\mathrm{comp}}(d)$ of Definition 4.11, whose gates may read raw coordinates and earlier gates together.

$\mathcal{A}_{\mathrm{lin}}(d) \subseteq \mathcal{A}^{\mathrm{thr}}_1(d) \subseteq \mathcal{A}^{\mathrm{thr}}_2(d) \subseteq \cdots$: a linear threshold is realized with one hidden unit, and a zero-weight hidden unit embeds $\mathcal{A}^{\mathrm{thr}}_m$ into $\mathcal{A}^{\mathrm{thr}}_{m+1}$; constants are in $\mathcal{A}_{\mathrm{lin}}$ ($w = 0$). Instance values used below: $\mathcal{A}_{\mathrm{lin}}(1)$: $K = 2$, $E = 3$; $\mathcal{A}^{\mathrm{thr}}_2(1)$: $K = 7$, $E = 11$; $\mathcal{A}^{\mathrm{thr}}_3(1)$: $K = 10$, $E = 16$; $\mathcal{A}_{\mathrm{lin}}(2)$: $K = 3$, $E = 5$; $\mathcal{A}^{\mathrm{thr}}_2(2)$: $K = 9$, $E = 15$.

Definition 4.2 (declared readout ladder)**.** A *declared readout ladder* is a finite chain $\mathcal{A}_1 \subseteq \cdots \subseteq \mathcal{A}_L$ of readout classes together with declared accounting values $K_1 < \cdots < K_L$ and $E_1 < \cdots < E_L$ under a stated convention, strictly increasing along the chain *by construction*.

The strict monotonicity of K and E along a ladder is part of the declaration, not a derived fact; the definition exists so that Section 4.4 can state exactly what is and is not claimed.

4.2 Family 1: XOR, and the gap closed at width 2

Lemma 4.3 (single-hidden-unit collapse)**.** *For every (Y, Z) and every d: $H_{\mathcal{A}_1^{\mathrm{thr}}(d)}(Y \mid Z) = H_{\mathcal{A}_{\mathrm{lin}}(d)}(Y \mid Z)$.*

Proof. Let $r \in \mathcal{A}_1^{\mathrm{thr}}(d)$, $r(z) = \mathbf{1}\{v_1 g_1(z) + c \geq 0\}$ with $g_1 \in \mathcal{A}_{\mathrm{lin}}(d)$. As $g_1(z) \in \{0, 1\}$, r is a constant, g_1, or $1 - g_1$. Constants give $H(Y \mid r(Z)) = H(Y)$, achievable in $\mathcal{A}_{\mathrm{lin}}$; $g_1 \in \mathcal{A}_{\mathrm{lin}}$; and $1 - g_1$ induces the same partition as g_1. So the infima over the two classes run over the same set of values. □

Theorem 4.4 (XOR gap, exact)**.** *Let $X = (X_1, X_2)$ uniform on $\{0, 1\}^2$, $Y = X_1 \oplus X_2$, $Z = X$, as in Proposition 3.6. Take $\mathcal{A}_1 = \mathcal{A}_{\mathrm{lin}}(2)$, $\mathcal{A}_2 = \mathcal{A}_2^{\mathrm{thr}}(2)$ (strict inclusion). Then*

$$H_{\mathcal{A}_1}(Y \mid Z) = \tfrac{3}{4} h_2\big(\tfrac{1}{3}\big) \approx 0.6887 \text{ bits},$$
$$H_{\mathcal{A}_2}(Y \mid Z) = 0,$$

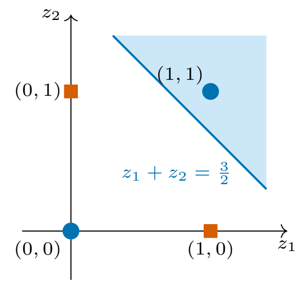
so the gap is exactly $\frac{3}{4} h_2(\frac{1}{3})$ bits, attained.

Proof. The first equality is Proposition 3.6. For the second, the two-hidden-unit threshold network with $g_1 = \mathbf{1}\{z_1 + z_2 \geq 1\}$, $g_2 = \mathbf{1}\{z_1 + z_2 \geq 2\}$, output $\mathbf{1}\{g_1 - g_2 - 1 \geq 0\}$, computes XOR exactly on $\{0, 1\}^2$: $(0, 0) \mapsto (0, 0) \mapsto 0$; $(0, 1), (1, 0) \mapsto (1, 0) \mapsto 1$; $(1, 1) \mapsto (1, 1) \mapsto 0$. Hence $H(Y \mid r(Z)) = 0$ for this $r \in \mathcal{A}_2$, and $H_{\mathcal{A}_2}(Y \mid Z) \geq H(Y \mid Z) = 0$ gives equality. □

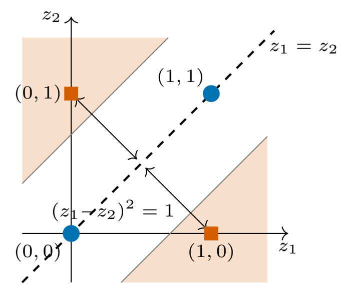
Figure 2 shows both halves of the XOR story: panel (a) the 3-vs-1 split behind the $\frac{3}{4} h_2(\frac{1}{3})$ floor, with the diagonal pair unreachable by any line; panel (b) the quadratic decoder of Proposition 4.8 (Section 4.5), which folds the plane about the diagonal and separates XOR with one threshold.

4.3 Family 2: sparse superposed features, with an equality rung

The family. d_{feat} binary features with exactly k active: F uniform on $\{f \in \{0, 1\}^{d_{\mathrm{feat}}} : \sum_j f_j = k\}$. A fixed projection $A \in \mathbb{R}^{d_z \times d_{\mathrm{feat}}}$ with $d_z < d_{\mathrm{feat}}$ superposes them, $Z = AF$, and the task reads one designated feature, $Y = F_{j^*}$. If $\mathrm{spark}(A) > 2k$ (every $2k$ columns linearly independent), then $F \mapsto AF$ is injective on k-sparse vectors [9], hence $H(F \mid Z) = 0$



(a) best halfspace: singleton $\{(1, 1)\}$ pure, triple with labels 2:1 (the 3-vs-1 split): $H = \frac{3}{4} h_2(\frac{1}{3}) \approx 0.6887$ bits (the diagonal pair is not linearly separable)



(b) the quadratic fold $r(z) = \mathbf{1}\{(z_1 - z_2)^2 \geq \frac{1}{2}\}$: both $Y{=}0$ points lie *on* the fold line, both $Y{=}1$ points at squared offset 1; the shaded region is $r = 1$, and $H(Y \mid r(Z)) = 0$

Figure 2: The XOR geometry. Circles: $Y = 0$; squares: $Y = 1$; all four points have mass $\frac{1}{4}$. (a) The exact linear-threshold floor of Proposition 3.6: the best achievable halfspace cell is a pure singleton against a 2:1 triple, giving $\frac{3}{4} h_2(\frac{1}{3}) \approx 0.6887$ bits, and the only partition that would do better (diagonal pair vs. diagonal pair) is exactly the non-achievable one. (b) The in-basis quadratic decoder of Proposition 4.8: $(z_1 - z_2)^2$ measures squared offset from the diagonal $z_1 = z_2$, collapsing each diagonal class to a single value (0 and 1), so one threshold at $\frac{1}{2}$ computes XOR exactly.

and a fortiori $H(Y \mid Z) = 0$: the representation is perfectly sufficient and all readout-constrained uncertainty is interface-induced cross-talk. (Efficient sparse recovery under incoherence is also classical [10], cited for context on achievability cost, not used in any proof.)

The minimal computed instance. $d_{\mathrm{feat}} = 4$, $k = 2$, $d_z = 1$, $A = (1\ 2\ 4\ 8)$, $Y = F_1$. The six equiprobable feature patterns give six distinct sums (atoms):

z	3	5	6	9	10	12
active pair	$\{1,2\}$	$\{1,3\}$	$\{2,3\}$	$\{1,4\}$	$\{2,4\}$	$\{3,4\}$
$y = f_1$	1	1	0	1	0	0

All sums distinct $\Rightarrow H(F \mid Z) = 0 \Rightarrow H(Y \mid Z) = 0$ (injectivity verified directly for this instance). The label pattern along the sorted atoms is $(1, 1, 0, 1, 0, 0)$: the $Y{=}1$ fiber is *interleaved* with the $Y{=}0$ fiber on the line, the one-dimensional analogue of the XOR diagonal, produced by superposition cross-talk rather than by parity. Figure 3 draws the instance: the interleaving, the best single threshold cut (the floor of part (ii) below), and the three cuts that decode exactly (part (iv)).

Theorem 4.5 (superposition gap, exact, with an equality rung)**.** *For the instance above:*

(i) $H(Y \mid Z) = 0$.

(ii) $H_{\mathcal{A}_{\mathrm{lin}}(1)}(Y \mid Z) = \frac{2}{3}h_2(\frac{1}{4}) = \frac{4}{3} - \frac{1}{2}\log_2 3 \approx 0.5409$ *bits.*

(iii) $H_{\mathcal{A}_2^{\mathrm{thr}}(1)}(Y \mid Z) = \frac{2}{3}h_2(\frac{1}{4})$ *as well: strict class enlargement with zero improvement.*

(iv) $H_{\mathcal{A}_3^{\mathrm{thr}}(1)}(Y \mid Z) = 0$*, attained.*

Proof. (i) Above.

(ii) A readout in $\mathcal{A}_{\mathrm{lin}}(1)$ partitions the line into a half-line and its complement; on six sorted atoms the achievable partitions are exactly the prefix/suffix splits of sizes $m = 0, \ldots, 6$ (complements give the same partition). With all atoms of mass $1/6$ and label pattern $(1, 1, 0, 1, 0, 0)$, the conditional entropies for $m = 0, \ldots, 6$ are

$$1, \quad \tfrac{5}{6}h_2(\tfrac{2}{5}), \quad \tfrac{2}{3}h_2(\tfrac{1}{4}), \quad h_2(\tfrac{1}{3}), \quad \tfrac{2}{3}h_2(\tfrac{3}{4}), \quad \tfrac{5}{6}h_2(\tfrac{3}{5}), \quad 1,$$

numerically 1, 0.8091, **0.5409**, 0.9183, **0.5409**, 0.8091, 1. The minimum is $\frac{2}{3}h_2(\frac{1}{4})$, attained at $m = 2$ (cell $\{3, 5\}$ is Y-pure; the complementary four atoms carry one $Y{=}1$) and at $m = 4$ (mirror). The class realizes only these finitely many partitions, so the infimum equals this minimum.

(iii) Upper bound by Proposition 4.1. Lower bound: let $r \in \mathcal{A}_2^{\mathrm{thr}}(1)$. The two hidden thresholds cut the line into at most three consecutive regions on which (g_1, g_2) is constant; the output assigns a binary label to each region, so $r^{-1}(1)$ is a union of at most two of three consecutive regions; hence one of the two cells is a single contiguous block of the sorted atoms (if a label gets two regions, they are either adjacent, merging into one block and leaving the other label one block, or they are the two outer regions, leaving the middle region as the other label's block). The achievable partitions are thus exactly the (contiguous block, complement) partitions. Enumerating all $\binom{6}{2} + 6 = 21$ blocks gives only four values of $H(Y \mid r(Z))$: 0.5409 (blocks $\{3, 5\}$, $\{3, 5, 6, 9\}$, $\{6, 9, 10, 12\}$, $\{10, 12\}$), 0.8091 (the six singletons and their mirrors), 0.9183, and 1. The minimum is again $\frac{2}{3}h_2(\frac{1}{4})$: isolating the stranded $Y{=}1$ atom at $z = 9$ together with $\{3, 5\}$ requires a cell with two separated blocks against a complement that is also disconnected (four label-runs along the line), which needs three cuts, not two. (The enumeration is finite and was additionally verified by exhaustive machine check over all cut-pairs and output labelings.)

(iv) Three hidden units suffice: $g_1 = \mathbf{1}\{-z \geq -5.5\}$ (i.e. $z \leq 5.5$), $g_2 = \mathbf{1}\{z \geq 7\}$, $g_3 = \mathbf{1}\{z \geq 9.5\}$, output $\mathbf{1}\{g_1 + g_2 - g_3 - 1 \geq 0\}$. At the atoms: $z = 3, 5 \mapsto (1, 0, 0) \mapsto 1$; $z = 6 \mapsto (0, 0, 0) \mapsto 0$; $z = 9 \mapsto (0, 1, 0) \mapsto 1$; $z = 10, 12 \mapsto (0, 1, 1) \mapsto 0$. This reproduces Y exactly, so $H(Y \mid r(Z)) = 0$. $\square$

One constructed instance thus simultaneously witnesses: a strict readout-class gap of exactly 0.5409 bits below a perfectly sufficient representation; an enlargement ($\mathcal{A}_1^{\mathrm{thr}} \to \mathcal{A}_2^{\mathrm{thr}}$, K_{unsh} from 2 to 7) that buys nothing: paying K without H improvement is possible, so the converse of the accounting clause is false and not claimed; and a sharp threshold ($m = 3$) at which the floor collapses to the Bayes value.

4.4 The accounting clause: what is proved, what is refuted, what is settled

The accounting claim attached to readout geometry (*strict improvement on geometry-dependent tasks is paid in readout K and/or E*) comes in pieces of decreasing strength that must not be conflated: a bookkeeping statement over declared ladders, a ladder-restricted theorem, two refuted universal forms, and a narrowed statement settled at the linear budget and closed at the frontier (Section 4.6).

Proposition 4.6 (ladder accounting: bookkeeping, stated as such)**.** *Let $\mathcal{A}_1 \subseteq \cdots \subseteq \mathcal{A}_L$ be a declared readout ladder (Definition 4.2). Then for every i: $H_{\mathcal{A}_{i+1}}(Y \mid Z) \leq H_{\mathcal{A}_i}(Y \mid Z)$ (Proposition 4.1), and every strict improvement $H_{\mathcal{A}_{i+1}} < H_{\mathcal{A}_i}$ coincides with $K_{i+1} > K_i$ and $E_{i+1} > E_i$.*

Proof. The first clause is Proposition 4.1. The second holds because $K_{i+1} > K_i$ and $E_{i+1} > E_i$ hold for *all* adjacent rungs by the ladder declaration, improvement or not. $\square$

Proposition 4.6 is bookkeeping over the declared ladder, not a law of nature (its mathematical content is exhausted by Proposition 4.1 plus the declaration),

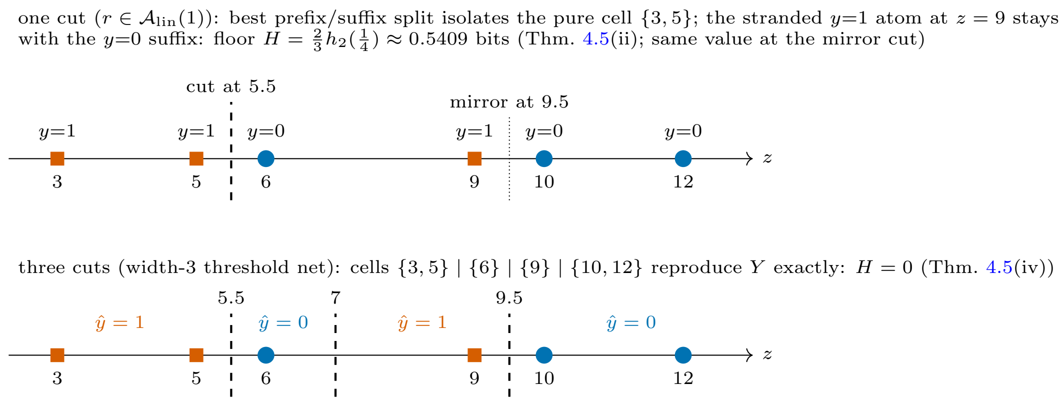


Figure 3: The superposition line instance: six equiprobable atoms $z \in \{3,5,6,9,10,12\}$ (sums of the active column pairs of $A = (1\ 2\ 4\ 8)$) with labels $(1,1,0,1,0,0)$ in sorted order: squares $y=1$, circles $y=0$. Top: any single linear threshold realizes only prefix/suffix splits; the best one pays $\frac{2}{3}h_2(\frac{1}{4}) \approx 0.5409$ bits because the y=1 fiber is interleaved. Two cuts provably do no better (Thm. 4.5(iii)). Bottom: three cuts (the width-3 network of Thm. 4.5(iv)) isolate the stranded atom at $z=9$ and decode exactly.

and is recorded because it is the exact form in which Experiment 2 checks the clause, stated so that no one can read the coincidence as *derived*.

Proposition 4.7 (payment floors for the hidden-layer threshold ladder)**.** *Under the Section 4.1 conventions:*

(1) (XOR family) Any $r \in \bigcup_m \mathcal{A}_m^{\mathrm{thr}}(2)$ with $H(Y \mid r(Z)) < \frac{3}{4}h_2(\frac{1}{3})$ for the Theorem 4.4 task has $m \geq 2$; hence $K_{\mathrm{unsh}} \geq 9$ and $E \geq 15$, versus $K_{\mathrm{unsh}} = 3$, $E = 5$ for $\mathcal{A}_{\mathrm{lin}}(2)$. Both bounds are attained (the Theorem 4.4 net achieves $H = 0$ at $m = 2$). Payment schedule for this family: floor 0.6887 bits at $(K_{\mathrm{unsh}}, E) = (3,5)$; floor 0 from $(K_{\mathrm{unsh}}, E) = (9,15)$.

(2) (Superposition instance) Any $r \in \bigcup_m \mathcal{A}_m^{\mathrm{thr}}(1)$ with $H(Y \mid r(Z)) < \frac{2}{3}h_2(\frac{1}{4})$ for the Theorem 4.5 task has $m \geq 3$; hence $K_{\mathrm{unsh}} \geq 10$ and $E \geq 16$, versus $K_{\mathrm{unsh}} = 2$, $E = 3$ for $\mathcal{A}_{\mathrm{lin}}(1)$. Attained (Theorem 4.5(iv)). Payment schedule: floor 0.5409 bits at $(K_{\mathrm{unsh}}, E) = (2,3)$, unchanged at $m = 2$, i.e. $(K_{\mathrm{unsh}}, E) = (7,11)$ (paying without improvement), and floor 0 from $(K_{\mathrm{unsh}}, E) = (10,16)$.

Proof. (1) $m \leq 1$ collapses to $\mathcal{A}_{\mathrm{lin}}(2)$ by Lemma 4.3, whose floor is $\frac{3}{4}h_2(\frac{1}{3})$ (Proposition 3.6). K/E values are the stated conventions at $m = 2$, $d = 2$. (2) $m \leq 1$ collapses to $\mathcal{A}_{\mathrm{lin}}(1)$ (Lemma 4.3, floor Theorem 4.5(ii)); $m = 2$ has the same floor by Theorem 4.5(iii); so beating it needs $m \geq 3$, and Theorem 4.5(iv) attains 0 at $m = 3$. K/E at $m = 3$, $d = 1$. □

Proposition 4.7 is a genuine (small) theorem, but it is restricted to the hidden-layer threshold ladder under the declared conventions: already within the declared operation basis, the strictly larger class of affine-threshold compositions (Definition 4.11) beats the XOR floor at $E = 12 < 15$ (Proposition 4.14; Remark 4.15). Outside that family, the corresponding *universal* claim (no readout at the linear budget beats a linear floor) is **false**, in two successively stronger senses, recorded next as propositions.

4.5 Two disproofs

The first disproof needs no exotic operations at all: it lives inside the declared $\{+, \times, \geq\}$ basis.

Proposition 4.8 (a quadratic decoder beats the XOR floor inside the declared basis)**.** *For the XOR task of Theorem 4.4, the two-weight, one-threshold readout*

$$r(z) = \mathbf{1}\{(z_1 - z_2)^2 \geq \tfrac{1}{2}\}$$

computes $Y = z_1 \oplus z_2$ exactly on $\{0,1\}^2$, so $H(Y \mid r(Z)) = 0$, beating the linear floor $\frac{3}{4}h_2(\frac{1}{3}) \approx 0.6887$ bits to zero. The readout is a threshold-of-arithmetic-circuit map over exactly the declared $\{+, \times, \geq\}$ basis (no transcendental operation), and its cost under the Section 4.1 conventions, fully parameterized as $\mathbf{1}\{(w \cdot z)^2 \geq \theta\}$ with $w = (1,-1)$, $\theta = \frac{1}{2}$, is $K = 3$ stored reals, equal to $K(\mathcal{A}_{\mathrm{lin}}(2)) = 3$, and $E = 6$ FLOPs (dot product 3, one squaring, threshold-subtract 1, compare 1) against $E(\mathcal{A}_{\mathrm{lin}}(2)) = 5$.

Proof. On $\{0,1\}^2$, $z_1 - z_2 \in \{-1, 0, 1\}$, so $(z_1 - z_2)^2 \in \{0, 1\}$, and $(z_1 - z_2)^2 = 1$ iff $z_1 \neq z_2$ iff $Y = 1$. Thresholding at $\frac{1}{2}$ therefore reproduces Y at all four points: $(0,0) \mapsto 0$, $(0,1) \mapsto 1$, $(1,0) \mapsto 1$, $(1,1) \mapsto 0$

(machine-verified by enumeration). The cost counts are the stated conventions applied to the circuit. ☐

Remark 4.9 (the one-FLOP artifact: the universal in-basis claim is dead either way). Under the fully-parameterized costing above, the quadratic decoder lands at $K = K(\mathcal{A}_{\text{lin}}(2))$ and $E = E(\mathcal{A}_{\text{lin}}(2)) + 1$: one squaring operation, with *zero* additional stored parameters, buys the entire 0.6887-bit floor down to zero. Whether this formally violates a "$K \leq$ and $E \leq$" clause depends on a costing convention at the granularity of a single FLOP: if the fixed coefficients $(1, -1)$ are treated as circuit constants rather than stored multiplicands (an equally defensible reading of the declared basis, under which the subcircuit $z_1 - z_2$ is a single $+$ operation), the cost is $K \leq 3$ (as low as 1: only θ stored) and $E \leq 4 < 5$, and the inequality is violated outright. A universal claim whose truth value flips on whether multiplication by a fixed ± 1 counts as a stored-parameter FLOP has no defensible content. We therefore record the basis-restricted universal statement as **refuted** for the XOR family and do not restate it. (For the superposition instance of Theorem 4.5 the same quadratic budget provably *ties* and does not beat the floor: a quadratic threshold on the line realizes exactly the interval/complement partitions of the six atoms, whose minimum is the linear floor $\frac{2}{3}h_2(\frac{1}{4})$, machine-verified by enumeration. The refutation is family-specific; the warning is not.)

Proposition 4.10 (the unrestricted-basis version is false: a sine decoder)**.** *Over an unrestricted operation basis, the universal no-free-improvement statement fails on the superposition family as well. For the Theorem 4.5 instance, the two-parameter readout*

$$r(z) = \mathbf{1}\{\sin(\omega z + \varphi) \geq 0\},$$
$$(\omega, \varphi) = (2.8616,\ 0.5236),$$

reproduces Y exactly at all six atoms: the sine values at $z = 3, 5, 6, 9, 10, 12$ are 0.311, 0.768, −0.915, 0.911, −0.761, −0.301 respectively (margin 0.30; machine-verified). Hence $H(Y \mid r(Z)) = 0$ with $K = 2 = K(\mathcal{A}_{\text{lin}}(1))$ and, if sin *is granted unit cost, $E \approx E(\mathcal{A}_{\text{lin}}(1))$.*

The sine construction is the classical infinite-VC shattering example [11] and can be deflected by refusing transcendental operations unit cost; Proposition 4.8 cannot (it uses only the declared operations) and is therefore the stronger warning. Together the disproofs say precisely where universal accounting claims die: once products of input-derived quantities are admitted, a single squaring folds the input space; with transcendental operations, one frequency parameter shatters everything. No budget-versus-floor law of the Proposition 4.7 type survives in either regime, except as an artifact of how the folding operation is costed.

4.6 The narrowed statement: settled at the linear budget, closed at the frontier

What remains after the disproofs is the statement restricted to the family in which the floors of this paper are actually *proved*: compositions of affine-threshold gates, where squaring of differences is outside the class by construction. We define the class, settle the linear-budget statement, pin the XOR family's minimal operation count exactly, close the superposition family's payment frontier completely, and then state what is genuinely open.

Definition 4.11 (affine-threshold compositions)**.** For $d \geq 1$ let $\mathcal{A}_{\text{comp}}(d)$ be the class of readouts computed by finite circuits whose gates are affine thresholds $u \mapsto \mathbf{1}\{w \cdot u + b \geq 0\}$, each gate's inputs u being coordinates of $z \in \mathbb{R}^d$ and/or outputs of earlier gates; coefficients are stored reals (a fan-in-f gate stores $f + 1$ reals, all counted by the unshared count K_{unsh}; the distinct-value count K_{val} counts numerically equal stored values once, Section 2.4); the operation basis acting on input-derived quantities is $\{+, \geq\}$: multiplication enters only through stored coefficients, so no product of two input-derived quantities (in particular no squaring of differences) is available in the class. Costing: a fan-in-f affine-threshold gate costs $2f + 1$ FLOPs, consistent with the Section 4.1 conventions, and E is the total per evaluation.

Proposition 4.12 (at the linear budget, affine-threshold compositions are linear thresholds)**.** *For the task families of Theorems 4.4 ($d = 2$) and 4.5 ($d = 1$): every $r \in \mathcal{A}_{\text{comp}}(d)$ with $E \leq E(\mathcal{A}_{\text{lin}}(d)) = 2d + 1$ induces the same partition of the support as some element of $\mathcal{A}_{\text{lin}}(d)$ (constants included, via $w = 0$). Hence $H(Y \mid r(Z)) \geq H_{\mathcal{A}_{\text{lin}}(d)}(Y \mid Z)$: no readout in the class at the linear E budget (regardless of K, and a fortiori under the joint budget $K \leq K(\mathcal{A}_{\text{lin}}(d))$, $E \leq E(\mathcal{A}_{\text{lin}}(d))$) achieves $H(Y \mid r(Z))$ strictly below the $\mathcal{A}_{\text{lin}}(d)$ floor.*

Proof. A fan-in-f gate costs $2f + 1 \geq 1$ FLOPs, so the multiset of gate fan-ins satisfies $\sum_g (2f_g + 1) \leq 2d + 1$. For $d = 2$ ($E \leq 5$) the possibilities are: a single fan-in-2 gate (cost 5), whose inputs must be coordinates because no earlier gate is affordable, hence an element of $\mathcal{A}_{\mathrm{lin}}(2)$; a single fan-in-1 gate (cost 3), possibly alongside fan-in-0 gates (cost 1 each, constants), where a fan-in-1 gate reading a coordinate is a linear threshold with one zero weight and a fan-in-1 gate reading a constant is itself constant; or fan-in-0 gates alone, i.e. constants. For $d = 1$ ($E \leq 3$): a single fan-in-1 gate reading z (an element of $\mathcal{A}_{\mathrm{lin}}(1)$), or constants alone. In every case the induced partition is realized in $\mathcal{A}_{\mathrm{lin}}(d)$, so the floor values of Proposition 3.6 and Theorem 4.5(ii) apply. (Machine check: exhaustive enumeration of all threshold-gate circuits on $\{0,1\}^2$ with $E \leq 5$, each gate ranging over all threshold functions of its inputs, produces exactly the 14 halfspace-achievable Boolean functions of Appendix A.1 and nothing else; Appendix A.3.) ☐

The apparent open content of such a conjecture (arbitrary depth and parameter sharing) is exactly what the E budget excludes: depth costs gates, every gate costs at least 3 FLOPs once it reads anything, and parameter sharing reduces K_{val} but never K_{unsh} or E. The linear-budget statement is therefore a citable cost-accounting fact, not a conjecture. What its resolution exposes is that the interesting quantity was never the linear budget itself but the *frontier*: how much E must be paid, at or near linear K, before a floor falls.

Definition 4.13 (payment frontier)**.** For a task family with linear floor $H_{\mathcal{A}_{\mathrm{lin}}}(Y \mid Z)$, the *payment frontier* of $\mathcal{A}_{\mathrm{comp}}$ is the budget set

$$\mathcal{F} = \big\{(K, E) : \exists\, r \in \mathcal{A}_{\mathrm{comp}} \text{ with } K(r) \leq K,\ E(r) \leq E,\ H(Y \mid r(Z)) < H_{\mathcal{A}_{\mathrm{lin}}}(Y \mid Z)\big\}.$$

$\mathcal{F}$ is a set of *budgets*, upward closed by construction: $(K, E) \in \mathcal{F}$ iff *some* circuit within the budget beats the floor, a point that matters below (Remark 4.21). $K(r)$ is counted under a declared stored-parameter convention; frontier statements in this paper use the unshared count K_{unsh} (Section 2.4) unless a quantity is explicitly marked K_{val}.

For the XOR family the frontier's minimal-E edge can be pinned exactly.

Proposition 4.14 (minimal operation count to beat the XOR floor)**.** *Within $\mathcal{A}_{\mathrm{comp}}(2)$, the minimum E at which any circuit achieves $H(Y \mid r(Z)) < \frac{3}{4} h_2(\frac{1}{3})$ for the XOR task of Theorem 4.4 is exactly $E = 12$, and the value achieved there is 0. Attainment: the depth-2 circuit*

$$g_1 = \mathbf{1}\{z_1 + z_2 - 2 \geq 0\} \text{ (fan-in 2, 5 FLOPs)},$$

$$r = \mathbf{1}\{z_1 + z_2 - 2g_1 - 1 \geq 0\} \text{ (fan-in 3, 7 FLOPs)}$$

computes XOR exactly, at $K_{\mathrm{unsh}} = 7$ stored reals (strictly inside the $(K_{\mathrm{unsh}}, E) = (9, 15)$ of the width-2 network of Theorem 4.4) and at $K_{\mathrm{val}} = 3 = K(\mathcal{A}_{\mathrm{lin}}(2))$ under the distinct-value convention (the equal stored values $\{1, -2, -1\}$ counted once).

Proof. Attainment: $(0,0) \mapsto g_1 = 0$, $r = \mathbf{1}\{-1 \geq 0\} = 0$; $(0,1),(1,0) \mapsto g_1 = 0$, $r = \mathbf{1}\{0 \geq 0\} = 1$; $(1,1) \mapsto g_1 = 1$, $r = \mathbf{1}\{-1 \geq 0\} = 0$: XOR exactly, so $H(Y \mid r(Z)) = 0$; the costs are the stated conventions (5+7 FLOPs, 3+4 stored reals). Minimality: a binary readout beats the $\frac{3}{4} h_2(\frac{1}{3})$ floor only by inducing a diagonal-pair partition (Appendix A.1: every other two-cell partition of the four points gives $\geq \frac{3}{4} h_2(\frac{1}{3})$), i.e. only by computing XOR or its complement; threshold functions are closed under complementation at equal cost, so it suffices that no circuit with $E \leq 11$ computes XOR. On the Boolean domain every gate computes a threshold function of its inputs, gates reading a repeated signal or a constant are dominated by cheaper gates of smaller fan-in, and at $E \leq 11$ at most three gates of fan-in at most 3 are affordable, so the search space is finite (threshold functions of 1, 2, 3 inputs number 4, 14, 104). Exhaustive machine enumeration over all gate fan-in multisets with $\sum_g (2f_g + 1) \leq 11$, all wirings, and all threshold functions per gate finds no circuit computing XOR; the enumeration's structure, the admissible circuit shapes, and the threshold-function censuses it uses are tabulated in Appendix A.3, and the committed script named there re-runs the whole search with hard assertions. (The instructive near-miss at $E = 10$: the shape $g_1 = \mathrm{threshold}(z_1, z_2)$, $r = \mathrm{threshold}(g_1, z_i)$ requires g_1 to separate both z_i-fibers, which forces $g_1 = \pm z_{3-i}$, and then r is a linear threshold of (z_1, z_2), blocked by Proposition 3.6.) ☐

Remark 4.15 ($E = 12$ does not contradict the ladder's $E \geq 15$ payment floor). Proposition 4.14's $E = 12$ and the hidden-layer threshold ladder's payment floor $E \geq 15$ for the same task (Proposition 4.7(1)) quantify over *different families*. In $\mathcal{A}_{\mathrm{comp}}(2)$ a gate may read raw coordinates and earlier gates together (the witness's output gate reads z_1, z_2, *and* g_1), and these skip connections are exactly what the ladder excludes: an $\mathcal{A}^{\mathrm{thr}}_m(2)$ output unit reads hidden units only (Section 4.1). The $E = 12$ witness is therefore a member of no $\mathcal{A}^{\mathrm{thr}}_m(2)$, and within the ladder $E = 15$ remains the exact floor.

The superposition edge: enumeration does not transfer, breakpoints do. The finite search behind Proposition 4.14 is Boolean: on $\{0,1\}^2$ every gate computes one of finitely many threshold functions, so circuits can be enumerated by truth table. The superposition instance of Theorem 4.5 lives on the real line (gates read a real-valued z, and the parameter space is a continuum), so the enumeration must be replaced. Three observations close the family completely: an exact entropy table over all binary partitions of the six atoms, a breakpoint-counting lemma for circuit outputs, and a finite enumeration over circuit *skeletons* (wiring shapes), which is finite even though the parameters are not. Every enumeration and case analysis cited below is re-run with hard assertions by the committed script `superposition_min_e.py` (Table 8), with all decision-relevant arithmetic exact (Appendix A.4).

Lemma 4.16 (breakpoint recursion)**.** *Call $h : \mathbb{R} \to \{0,1\}$ piecewise constant with B_h breakpoints if B_h is the number of points at which it changes value. Let g be an affine-threshold gate in $\mathcal{A}_{\text{comp}}(1)$ with gate inputs $h_1, \dots, h_j$ (each piecewise constant with B_{h_i} breakpoints) and z-weight w_z, and set $S = \sum_i B_{h_i}$. Then the output of g is piecewise constant with*

$$B_g \;\le\; \begin{cases} 2S+1, & w_z \neq 0 \ (\textit{the gate reads } z), \\ S, & w_z = 0. \end{cases}$$

Proof. The inputs' breakpoints cut $\mathbb{R}$ into at most $S+1$ open intervals on which every h_i is constant, so on each such interval the pre-activation $w_z z + \sum_i v_i h_i(z) + b$ is affine in z. If $w_z = 0$ it is constant there: g is constant per interval, and its breakpoints lie among the S cut points. If $w_z \neq 0$ it is strictly monotone there, so $\mathbf{1}\{\cdot \ge 0\}$ changes value at most once per interval; adding the at most S interval boundaries gives $B_g \le (S+1) + S = 2S+1$. ☐

Remark 4.17 (the naive count is false, and the gap is the mechanism). One might hope for the simpler count "a circuit containing t thresholds that read z is piecewise constant on at most $t+1$ intervals." That is true when no gate reads another gate (a width-m network's hidden layer places m cut points, giving at most $m+1$ regions, which is exactly the interval argument in the proof of Theorem 4.5(iii)) and *false* in general: a gate reading z *and* earlier gates is affine in z on each level region of its gate inputs and can cross zero once *per region*, so two thresholds reading z can produce *three* output breakpoints. This is not a defect of the lemma; it is the mechanism the minimal witness below exploits: the same skip-connection trick as the XOR $E = 12$ witness of Proposition 4.14 (Remark 4.15). The width-only special case is recovered as the $w_z = 0$ branch.

Proposition 4.18 (breakpoint budgets price the superposition floor)**.** *For the six-atom instance of Theorem 4.5, grade each two-cell partition P of the atoms by its* breakpoint count *$b(P)$: the number of adjacent pairs along the sorted atoms $3 < 5 < 6 < 9 < 10 < 12$ that P separates; equivalently, the minimal number of breakpoints of any piecewise-constant $r : \mathbb{R} \to \{0,1\}$ inducing P. Then:*

- *(i) Over all $2^6/2 = 32$ partitions, $H(Y \mid r(Z))$ takes exactly five values: 0, the linear floor $\frac{2}{3}h_2(\frac{1}{4}) \approx 0.5409$, $\frac{5}{6}h_2(\frac{2}{5}) \approx 0.8091$, $h_2(\frac{1}{3}) \approx 0.9183$, and 1. The only partition strictly below the floor is the exact-decode partition $\{3,5,9\}$ vs. $\{6,10,12\}$ (label pattern 110100 along the line), with $H = 0$ and $b = 3$: for this instance, beating the floor at all and decoding exactly are the same problem.*
- *(ii) The minimum of $H(Y \mid r(Z))$ over partitions with $b \le 2$ equals the linear floor $\frac{2}{3}h_2(\frac{1}{4})$ exactly: a second breakpoint buys nothing.*
- *(iii) At $b = 3$ the minimum is 0, attained by the exact-decode partition and only by it.*

Proof. (i) is a finite computation, machine-checked in exact arithmetic: the entropies of the 32 partitions are computed as rational coefficient vectors over the basis $(1, \log_2 3, \log_2 5)$, which is linearly independent over $\mathbb{Q}$ (a relation $2^a 3^b 5^c = 1$ with rational exponents reduces, clearing denominators, to an integer one, which forces $a = b = c = 0$ by unique factorization), so equality and order of entropy values are decided exactly; the five values and the uniqueness of the sub-floor partition are asserted by the committed script. *(ii)* A two-cell partition with $b \le 2$ has at most three label runs along the line, so one of its cells is a single contiguous block of the sorted atoms; the $b \le 2$ minimum is therefore the minimum over the $\binom{6}{2} + 6 = 21$ block/complement partitions already enumerated in the proof of Theorem 4.5(iii), namely the floor. (The best pattern with b exactly 2, an interior singleton, gives $\frac{5}{6}h_2(\frac{2}{5}) > \frac{2}{3}h_2(\frac{1}{4})$; the block optimum is a prefix split with $b = 1$.) *(iii)* The label pattern 110100 switches membership exactly three times, so the exact-decode partition has $b = 3$; (i) gives the value and the uniqueness. ☐

Proposition 4.19 (minimal operation count to beat the superposition floor)**.** *Within $\mathcal{A}_{\text{comp}}(1)$, the minimum E at which any circuit achieves $H(Y \mid r(Z)) <$*

$\frac{2}{3}h_2(\frac{1}{4})$ *for the Theorem 4.5 task is exactly* $E = 8$*, and the value achieved there is* 0*: beating the floor and exact decode coincide by Proposition 4.18(i). Attainment: the depth-2 skip-connection circuit*

$$g_1 = \mathbf{1}\{z - 7.5 \ge 0\} \text{ (fan-in 1, 3 FLOPs)},$$
$$r = \mathbf{1}\{-z + 4g_1 + 5.5 \ge 0\} \text{ (fan-in 2, 5 FLOPs)}$$

computes Y *exactly at all six atoms, at* $E = 3+5 = 8$ *and* $K_{\text{unsh}} = 2+3 = 5$*. At* $E = 8$ *the circuit shape is forced, so* $K_{\text{unsh}} = 5$ *is forced; the minimal distinct-value count at* $E = 8$ *is exactly* $K_{\text{val}} = 4$*, attained by* $g_1 = \mathbf{1}\{-z + 7 \ge 0\}$*,* $r = \mathbf{1}\{-z - 4g_1 + 9.5 \ge 0\}$ *(stored values* $\{-1, 7, -1, -4, 9.5\}$*: four distinct), and* $K_{\text{val}} = 3$ *is infeasible.*

Proof. Attainment. At the atoms: $z = 3, 5 \mapsto g_1 = 0$ and $-z + 5.5 = 2.5, 0.5 \ge 0 \mapsto 1$; $z = 6 \mapsto g_1 = 0$, $-0.5 \mapsto 0$; $z = 9 \mapsto g_1 = 1$, $-9 + 4 + 5.5 = 0.5 \mapsto 1$; $z = 10, 12 \mapsto g_1 = 1$, $-0.5, -2.5 \mapsto 0$: the label pattern 110100 exactly, so $H(Y \mid r(Z)) = 0$ (verified at all six atoms in exact rational arithmetic by the committed script, as is the second witness). Its three breakpoints 5.5, 7.5, 9.5 place one in each required gap and meet the Lemma 4.16 bound $2 \cdot 1 + 1 = 3$ with equality.

Minimality. Reduce first: every circuit has a reduced instance of no greater cost inducing the same output partition, where *reduced* means no fan-in-0 gates (a constant input is absorbed into the reading gate's bias, lowering that gate's fan-in and cost), no zero-weight inputs (dropped), and no unread non-output gates (deleted). A reduced gate has fan-in ≥ 1, hence cost ≥ 3. By Proposition 4.18, a beating circuit's output has at least 3 breakpoints. A single reduced gate at $d = 1$ reads z alone (the input layer of the line has fan-in 1) and has $B \le 1$, so $n \ge 2$ gates are needed. With $n = 2$ there are two reduced shapes: $g_1 \leftarrow z$, $r \leftarrow \{g_1\}$, whose output has $B \le S = 1$ (Lemma 4.16, $w_z = 0$ branch); and $g_1 \leftarrow z$, $r \leftarrow \{z, g_1\}$, the witness shape, with $E = 3 + 5 = 8$ and $K_{\text{unsh}} = (1+1) + (2+1) = 5$ forced. With $n \ge 3$, $E = \sum_g (2f_g + 1) \ge 3n \ge 9$. Hence no circuit with $E \le 7$ beats the floor (indeed, by Proposition 4.18(i), none achieves $H < \frac{2}{3}h_2(\frac{1}{4})$ at all), and $E = 8$ admits exactly one shape. (Machine check: exhaustive enumeration of all 49 reduced skeletons with $E \le 16$ confirms that every $E \le 7$ skeleton has output breakpoint bound ≤ 1 and that the unique $E = 8$ skeleton admitting 3 breakpoints is the witness shape; Appendix A.4.)

Value sharing. At $E = 8$ the forced form is $g_1 = \mathbf{1}\{w_1 z + b_1 \ge 0\}$, $r = \mathbf{1}\{wz + vg_1 + b \ge 0\}$ with $w_1, w, v \ne 0$: five stored reals. Placing one output breakpoint in each of the gaps $(5, 6)$, $(6, 9)$, $(9, 10)$ forces g_1's threshold into the middle gap (machine check: the split $\{3, 5, 6\}$ vs. $\{9, 10, 12\}$ is the unique one whose two region patterns are both monotone with a common slope sign) and pins r's two region crossings to the outer gaps, determining b and v from the crossing positions. In each of the four sign cases of $(\operatorname{sign} w_1, \operatorname{sign} w)$, every pattern of value coincidences that would leave at most three distinct stored values ($K_{\text{val}} \le 3$) reduces to an interval condition on the threshold and crossing positions that is empty over their admissible ranges (all twelve case–condition pairs machine-checked, boundary contacts included). So $K_{\text{val}} = 4$ is necessary, and the second witness attains it. ☐

Proposition 4.20 (the superposition payment frontier, completely characterized)**.** *For the Theorem 4.5 instance, with* K_{unsh} *the unshared count,*

$$\mathcal{F} \;=\; \{(K_{\text{unsh}}, E) : K_{\text{unsh}} \ge 5,\;\; E \ge 8\},$$

and both edges are attained simultaneously: the Proposition 4.19 witness sits at the corner $(5, 8)$*.*

Proof. By the minimality argument of Proposition 4.19, every beating circuit reduces (at no increase in K or E) to one with $n \ge 2$ gates and total fan-in $\sum_g f_g \ge 3$: the only $n = 2$ beating shape has fan-ins $\{1, 2\}$, and for $n \ge 3$ each reduced gate has $f_g \ge 1$. Hence every beating circuit has $K_{\text{unsh}} = \sum_g (f_g + 1) = \sum_g f_g + n \ge 5$ and $E = \sum_g (2f_g + 1) = 2\sum_g f_g + n \ge 8$. The corner is attained by the witness, and $\mathcal{F}$ is upward closed (Definition 4.13). ☐

In the two conventions at once: for the superposition family, the unshared-count frontier is exactly $K_{\text{unsh}} \ge 5$, $E \ge 8$. At the corner the distinct-value count is $K_{\text{val}} = 4$; the $K_{\text{val}} \le 3$ edge above $E = 8$ remains open (Open Problem 4.22(i)).

Remark 4.21 (a correction: budgets are not shapes)*.* An earlier version of this paper recorded the budget point $(7, 11)$ (the cost of the width-2 network shape) as provably outside $\mathcal{F}$, conflating a parametric *shape* with the *budget* it occupies: since $(5, 8) \in \mathcal{F}$ and $\mathcal{F}$ is upward closed, $(7, 11)$ *is* in $\mathcal{F}$, via the cheaper skip-connection witness strictly inside it. Theorem 4.5(iii) stands unchanged (no width-2 *network* beats the floor at any parameter setting), and the corrected gloss is itself the useful caution: an impossibility proved for a shape does not transfer to the budget it occupies.

Against the XOR family the comparison is instructive: the superposition edge is *lower* $(8 < 12)$ even though the task needed a strictly larger rung of the width-graded ladder ($m = 3$ vs. $m = 2$, Proposition 4.7). Two structural reasons, both visible in the proofs: on the line the entry-level gate costs 3 FLOPs rather than 5 (input fan-in is 1), and the skip connection's breakpoint amplification ($B \leq 2S+1$, Lemma 4.16) is at its most favorable when a single cheap gate already supplies the level structure. Both minimal witnesses, XOR's $E = 12$ circuit and the $E = 8$ circuit here, are instances of the same trick: the output gate reads the raw input alongside the hidden gate.

Open Problem 4.22 (the frontier residue)**.** *What remains genuinely open after Propositions 4.14–4.20.* (i) The $K_{\text{val}} \leq 3$ edge above the corner: *at $E = 8$ the distinct-value minimum is exactly $K_{\text{val}} = 4$ (Proposition 4.19); whether $K_{\text{val}} \leq 3$ becomes feasible at some $E > 8$ (more gates, more opportunities for value coincidence) is open, the Proposition 4.19 case analysis being specific to the forced two-gate form, as is the XOR family's minimal K_{unsh} at each admissible E (under the distinct-value convention the Proposition 4.14 witness already sits at $K_{\text{val}} = 3 = K(\mathcal{A}_{\text{lin}}(2))$, so for XOR value sharing makes the stored-parameter count non-binding).* (ii) General finite atom geometries on the line: *the reduction (exact entropy table, breakpoint recursion, finite skeleton enumeration) applies verbatim, but the minimal E as a function of the label pattern's required breakpoint count b is open. Lemma 4.16 permits breakpoint counts that grow exponentially in the number of chained skip gates, suggesting $\Theta(\log b)$ gates beyond the entry gate; but the lemma bounds counts, not placements, and realizability of arbitrary breakpoint* placements *at that cost is exactly what would have to be proved.* (iii) $d > 1$: *the breakpoint argument is one-dimensional; for multi-dimensional superposition instances hyperplane-arrangement cell counts replace interval counts, and nothing here settles those edges.*

A scope note on the accounting convention: both stored-parameter frontiers (K_{unsh} and K_{val}) count stored real values only; neither counts circuit topology or program description length. A description-length accounting would be a different convention and may move either edge.

Figure 4 plots every (K_{unsh}, E) point this paper proves something about, for each family: the linear budgets where the floors bind, the attaining circuits, the XOR family's exact minimal-E edge at $E = 12$, with that panel's unresolved K_{unsh} structure (Open Problem 4.22(i)) left visibly blank, and the superposition family's frontier, now fully two-toned: every budget is either excluded or in $\mathcal{F}$ (Proposition 4.20).

Within affine-threshold compositions, floors are governed by the partition geometry of hyperplane (or cut-point) arrangements, and a K/E budget translates into a counted bound on achievable cells (intervals on the line, with skip connections amplifying them at exactly the Lemma 4.16 rate), which is what makes the floors and Propositions 4.12–4.20 provable; outside the class, Propositions 4.8 and 4.10 show that no statement of this shape survives at all. The narrowed statement is where the theorem actually lives: at the linear budget as accounting, at the frontier as breakpoint geometry, with the open mathematics concentrated in the residue (Open Problem 4.22).

5 Experiments

Three experiments test the theory where ground truth is exact, then carry the central finding to real images and learned representations. Experiment 1 (Section 5.1) is the protocol's analytic calibration: a constructed Gaussian channel on which every prediction is exactly computable, and on which the masking signature is first exhibited. Experiment 2 (Section 5.2) demonstrates the floor/ladder accounting empirically: trained readouts land on their exactly computed floors, and the only readout that beats a floor pays the counted price. Experiment 3 (Section 5.3) is the headline: the masking signature survives real images and end-to-end learning. Per-experiment full tables and secondary observations are in Appendix E.

How the claims were protected. For each experiment, the generative construction, seeds, training recipes, analytic predictions, and pass/fail thresholds with tolerances were committed to a version-controlled archive before any run; each analysis ran exactly once on the committed protocol; and every outcome is reported, including a failed prediction (Section 5.3.1) and a criterion that passed its point estimates but is graded inconclusive by its own statistical-power provision (Appendix B). Experiments 1 and 2 ran with no changes to their committed protocols; in Experiment 3 one validation step in the committed protocol was corrected before any outcome was computed (Appendix B). Where a results table cites a criterion by its short archive label (Table 2), the row also states in words what the criterion required. The full protocol, criterion wordings, and amendment record are in Appendix B; per-experiment measurement details (estimator-bias accounting, fold schemes, measurement-quality checks) are in Appendix C.

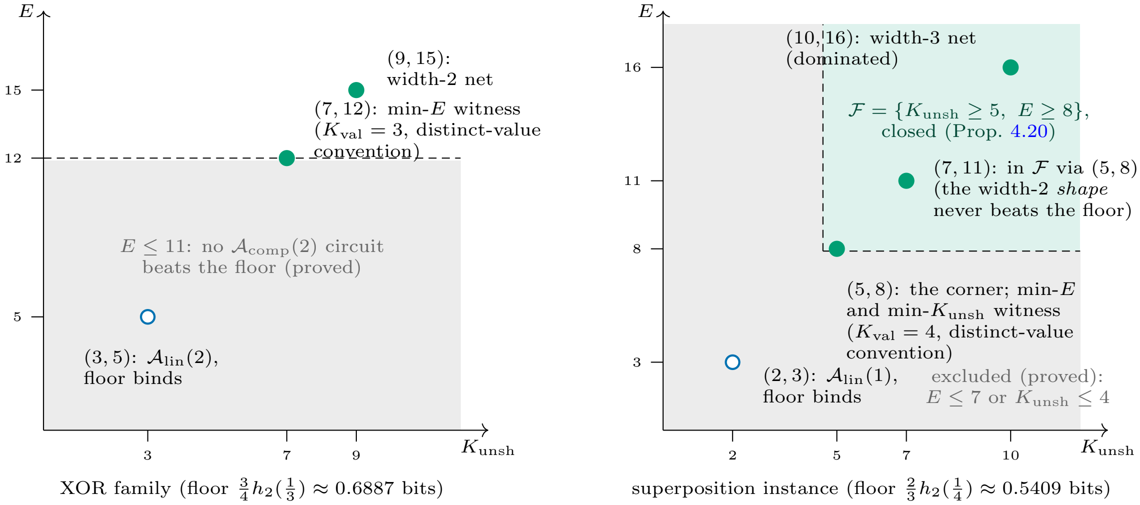


Figure 4: The payment frontier (Definition 4.13), one panel per family; axes are unshared stored-parameter counts K_{unsh} (Section 2.4) and per-evaluation operations E under the Section 4.1 conventions; the parenthetical K_{val} notes report the same witnesses under the different, distinct-value convention. Filled circles: budgets at which an affine-threshold composition provably beats the linear floor to $H = 0$ (Thm. 4.4, Props. 4.14 and 4.19, Thm. 4.5(iv)); open circles: budgets at which the floor provably binds (Prop. 4.12). Left: for XOR the shaded region $E \le 11$ is exhaustively excluded and the minimal-E edge sits exactly at $E = 12$ (Prop. 4.14); the $E = 15$ point is the hidden-layer threshold-ladder witness, the $E = 12$ point cheaper because $\mathcal{A}_{\text{comp}}(2)$ allows the second gate to read both raw coordinates and the earlier gate's output (Remark 4.15); the K_{unsh} structure off the known points belongs to the residue (Open Problem 4.22(i)). Right: the superposition panel is closed (Prop. 4.20): the frontier is exactly the quadrant $\mathcal{F} = \{K_{\text{unsh}} \ge 5,\ E \ge 8\}$, its corner attained by the Prop. 4.19 witness, every budget outside it excluded, and the budget point $(7, 11)$ in $\mathcal{F}$ even though the width-2 network shape priced there never beats the floor (Thm. 4.5(iii), Remark 4.21).

5.1 Experiment 1: analytic calibration on a constructed channel

Experiment 1's role is calibration: every quantity the protocol measures is exactly computable from the construction, so the estimators, tolerances, and frozen-readout shift protocol are validated against analytic ground truth before anything is measured where ground truth is unavailable. The masking signature is also first exhibited here in its cleanest form.

Setup and protocol. A four-dimensional Gaussian signal $S \sim \mathcal{N}(0, I_4)$ drives a binary task, $Y \mid S \sim \text{Bernoulli}(\sigma(w \cdot S))$ with fixed $w = (1, 1, 1, 1)$; writing $u = w \cdot S/\|w\| \sim \mathcal{N}(0, 1)$, the task logit is $2u$. Alongside ride $d_n \in \{0, 8, 64, 256\}$ nuisance coordinates, each correlated with the signal but causally useless ($N_j = \rho u + \sqrt{1-\rho^2}\,\varepsilon_j$ with $\varepsilon_j \sim \mathcal{N}(0, 1)$ i.i.d., $\rho_{\text{train}} = 0.9$); the shift flips the coupling sign, $\rho_{\text{shift}} = -0.9$, leaving every marginal unchanged. Three representations are compared: the full input X; the lawful compression S (sufficient by construction); and the unlawful compression N (insufficient by construction, since $\text{Var}(u \mid N) > 0$ exactly). The readout is logistic regression trained by damped Newton (fixed recipe), 10 seeds, $n_{\text{train}} = 50{,}000$, $n_{\text{test}} = 100{,}000$; every cross-entropy is measured out-of-sample. The analytic value of every in-distribution cross-entropy (Gauss–Hermite quadrature), the post-shift predictions for the frozen readout, and the seed-mean tolerance (0.010 bits, sized to the predicted estimator bias; Appendix C) were committed before any sample was drawn; the analysis ran once; all cells are reported. For the X and S arms the task is logistic-linear by construction, so $\text{CE}^{\star}_{\mathcal{A}} = H(Y \mid Z)$ exactly. Post-shift measurements estimate no entropy and are reported as $\widehat{\text{CE}}_Q(r_P)$ (Definition 2.2).

Calibration outcome: measured equals predicted. All five committed criteria passed; the full per-cell table is Table 9 (Appendix E). Lawful compression was measured to be free: the compressed arm S matched the full arm X within $+0.00000$ to $+0.00395$ bits across $d_n = 0$–256 (an order of magnitude or more inside the tolerance), at counted reductions of up to $57.9\times$ in parameters ($521 \to 9$), $1397\times$ in training FLOPs, and $35\times$ in inference FLOPs, with measured training FLOPs equal to the committed formula values exactly. By Corollary 3.8 in its sign-robust form, the substitution $X \to S$ strictly reduces $K + H_{\mathcal{T}} + \lambda E$ for every $\lambda \ge 0$; no multiplier value was used or tuned. And the measurement landed on analytic ground truth: every (arm, d_n) cell's measured

$\widehat{\mathrm{CE}}_{\mathrm{ID}}$ fell within 0.0035 bits of its committed analytic value, the worst deviations matching the predicted maximum-likelihood bias (0.0038 bits) almost exactly. The estimator's measured bias is the predicted bias, which is what licenses the same estimator and tolerances at scale.

The masking signature, first exhibition. The unlawful arm carries the finding (Table 1). The construction makes the N arm's insufficiency *analytically* nearly invisible in-distribution at large d_n: the true excess $H(Y \mid N) - H(Y \mid S)$ is 0.0123, 0.0016, 0.0004 bits at $d_n = 8, 64, 256$ (the latter two below the proxy resolution δ), and the measured excesses track these values (as pre-registered, no measured in-distribution strict-increase claim is made at $d_n \geq 64$). Under the committed sign-flip shift, the frozen readout's transfer cross-entropy $\widehat{\mathrm{CE}}_Q(r_P)$ on the same arm degrades by 1.6666, 1.7311, 1.7555 bits relative to its in-distribution value, landing within 0.0048, 0.0051, 0.0140 bits of the analytic predictions committed before the run; the lawful arm, required to be invariant under the same shift, moved at most 0.002 bits at every d_n. Figure 5 shows the asymmetry at a glance: the insufficient arm's in-distribution excess and the lawful arm's post-shift drift are pinned to the axis, while the same insufficient arm's post-shift elevation of $\mathrm{CE}_Q(r_P)$ towers at ≈ 1.7 bits, above the 1-bit line that no entropy difference on a binary task can cross.

What kind of quantity is the 1.7-bit degradation? It is *not* a conditional entropy: the post-shift values (2.34–2.43 bits) exceed the 1-bit entropy bound of a binary task, which is possible precisely because $\mathrm{CE}_Q(r_P)$ is the cross-entropy of a P-fit readout under Q (Definition 2.2), not an entropy. Nor is it information loss: the sign flip $\rho \mapsto -\rho$ leaves every marginal (and $I(N; Y)$) essentially unchanged, and a readout *retrained under* Q would land back at the in-distribution numbers, showing none of the degradation. The 1.7 bits is frozen-readout **transfer risk**: the P-fit interface points the wrong way under Q. An insufficiency whose in-distribution footprint is 0.0004 bits (unmeasurable under any realistic tolerance) thus manifests as deployment brittleness that is structurally invisible to any in-distribution evaluation and to any evaluation that is allowed to refit; an audit that evaluates only in-distribution, or that silently refits, can certify a representation carrying an arbitrarily large latent deployment liability. The committed failure conditions did not occur at any cell, and the committed measurement-quality checks all passed; pre-declared secondary observations (including the raw X arm's own drift under shift) are in Appendix E.

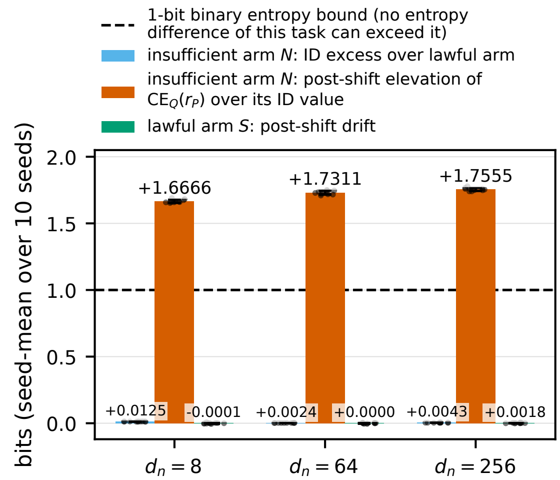


Figure 5: Experiment 1 headline result, computed from the committed raw per-seed results file: for each d_n, the insufficient arm's in-distribution excess held-out cross-entropy over the lawful arm (left bars, near zero: the masking), the same arm's post-shift elevation of the frozen-readout transfer cross-entropy $\mathrm{CE}_Q(r_P)$ over its own in-distribution value (center bars, ≈ 1.7 bits), and the lawful arm's post-shift drift (right bars, near zero). Bars are seed-means over the 10 paired seeds, whiskers one cross-seed standard deviation, dots the per-seed values. The dashed line marks the 1-bit binary entropy bound: the elevation exceeds it because $\mathrm{CE}_Q(r_P)$ is the cross-entropy of a P-fit interface under Q (Definition 2.2), not an entropy.

5.2 Experiment 2: the readout ladder

Setup and protocol. The construction is a scaled-up sibling of the superposition family of Section 4.3. A feature vector F is uniform on the $\binom{16}{4} = 1820$ binary patterns with exactly $k = 4$ of $d_{\mathrm{feat}} = 16$ features active; a fixed per-seed Gaussian projection superposes them, $Z = AF$, noiseless; the task reads one feature, $Y = F_1$. Atom injectivity was verified per cell before the run, so $H(Y \mid Z) = 0$ *exactly* everywhere: every bit of measured uncertainty is interface-induced. Superposition load $d_{\mathrm{feat}}/d_z = 4.0$–$1.33$ across $d_z \in \{4, 6, 8, 12\}$; 10 seeds; $n_{\mathrm{train}} = 20{,}000$, $n_{\mathrm{test}} = 100{,}000$. Three readout rungs of strictly increasing declared cost (Table 11, Appendix E): (i) a soft linear-logistic readout, whose population optimum is $\mathrm{CE}^{\star}_{\mathrm{lin}}$ (Definition 2.3); (ii) a two-layer ReLU network of width 64; (iii) an oracle nearest-atom decoder over the 1820 constructed atoms (a codebook-lookup *accounting* rung, not a trained readout). The key design choice: the linear class's floor is an *exact population quantity*, computed per seed by convex optimization on the exact 1820-atom distribution

Table 1: Experiment 1, shift protocol: pre-registered prediction vs. measurement (seed-means, 10 seeds, bits). Post-shift values are frozen-readout transfer cross-entropies $\widehat{\mathrm{CE}}_Q(r_P)$ (Definition 2.2), not conditional entropies; they may and do exceed the 1-bit binary entropy bound. "Lawful drift" is $\widehat{\mathrm{CE}}_{\mathrm{shift}}(S) - \widehat{\mathrm{CE}}_{\mathrm{ID}}(S)$; the analytic in-distribution excess of the unlawful arm over the lawful arm is shown for the masking comparison.

	ID excess, N arm		$\widehat{\mathrm{CE}}_Q(r_P)$, N arm			lawful drift
d_n	analytic	meas.	pred.	meas.	|dev.|	meas.
8	0.0123	+0.0125	2.3414	2.3463	0.0048	−0.00012
64	0.0016	+0.0024	2.4047	2.3996	0.0051	+0.00003
256	0.0004	+0.0043	2.4118	2.4259	0.0140	+0.00184

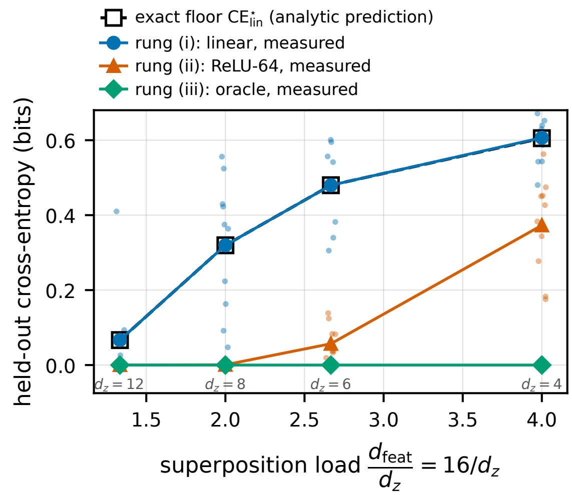


Figure 6: Experiment 2 ladder, computed from the committed raw per-seed results file: measured held-out cross-entropy of the three rungs versus superposition load d_{feat}/d_z, with the exact per-seed population floors $\mathrm{CE}^{\star}_{\mathrm{lin}}$ (computed by convex optimization and committed before the run; the file's `analytic_floor` column) overlaid as the dashed prediction line. Solid markers are seed-means over 10 seeds; faint dots are the per-seed values, whose vertical spread is real cross-projection heterogeneity (each seed draws its own A), not measurement noise. The measured linear rung is visually indistinguishable from its committed floor (worst seed-mean deviation 0.0013 bits); the oracle rung sits at the clip artifact 1.44×10^{-6} bits at every load.

and committed before the run; no training budget moves a linear readout below it, which is what makes "equivalent-budget comparison" exact rather than rhetorical. Committed tolerance 0.010 bits, with a statistical-power provision grading any comparison whose cross-seed standard error exceeds 0.005 bits inconclusive; the analysis ran once; all cells and seeds are reported.

Results. All primary criteria passed (Table 10, Appendix E); Figure 6 plots the whole experiment as prediction against measurement. The exact floors are real, out-of-sample: the trained linear rung reproduced its committed per-seed $\mathrm{CE}^{\star}_{\mathrm{lin}}$ floor with seed-mean deviations +0.0013, −0.0001, +0.0000, +0.0002 bits at $d_z = 4, 6, 8, 12$. Beating a floor carried the counted price: at the declared gap cells $d_z \in \{4, 6, 8\}$ the oracle rung achieved 1.44×10^{-6} bits (exactly the pre-registered probability-clip artifact; decode accuracy 1.0), beating the exact linear floors by 0.6053, 0.4799, 0.3196 bits (below a floor no training budget within the linear class can cross), carried by a rung whose counted K is 1820× larger and whose counted inference cost is 1577–1978× larger. (The oracle rung proves that zero task uncertainty is *available* given codebook storage and search; it does not prove that ordinary training finds it cheaply.) The accounting held everywhere: K and E strictly increase up-ladder at all 40 cells exactly as committed (Proposition 4.6), and the committed failure condition (a readout at linear K/E measuring below the exact linear floor by more than the tolerance) did not occur at any cell or seed. At $d_z = 12$, where 5 of 10 seeds are linearly separable (floor ≈ 0), rungs (i) and (ii) both sit at their floors: no superposition pressure, no gap, the data-side analogue of the equality rung of Theorem 4.5(iii). No measured quantity here estimates a hard-class $H_{\mathcal{A}}$: the theorem-side hard floors of Theorem 4.5 and the experiment-side soft floors are different objects, each computed exactly in its own way.

A committed secondary criterion (whether the *trained* width-64 network beats the exact linear floor by pre-set margins) passed its point estimates at both declared cells, yet is reported inconclusive because the registration's own statistical-power provision binds; the full episode, a worked example of the protocol applied against the experimenters' reading of the evidence, is in Appendix B.4. Measurement-quality checks and one infrastructure disclosure are in Appendix C.

5.3 Experiment 3: masking at scale

Experiments 1 and 2 run where ground truth is analytic; Experiment 3 scales the masking finding to real images and *learned* representations, with the construction retreating to the one place it is still needed: the spurious coupling itself. The question, committed before any run: does the masking signature (a cue-reliant representation indistinguishable from or better than a sufficient one in-distribution, whose frozen interface collapses under a coupling-breaking shift) reproduce when the representation is a CNN trained end-to-end on CIFAR-10?

Setup. Every train, test, and shift image set is built by one pinned construction module and no other code (Appendix C). CIFAR-10 (50,000 train / 10,000 test, $|\mathcal{Y}| = 10$ exactly uniform, so $H(Y) = \log_2 10 \approx 3.3219$ bits); per image, an indicator $s \in \{0, \ldots, 9\}$ drawn with $P(s = y) = \rho$ (off-diagonal uniform), and a per-class chromatic tint (ten maximally separated zero-mean hues, distinct, hence injective in s) added at fixed strength 0.20 (Figure 7). Two arms differ *only* in the coupling: the **naive** arm trains at $\rho \in \{0.7, 0.9, 0.95, 0.99\}$ (the tint is informative about y only through the train-time coupling: insufficiency by construction); the **lawful** arm trains decorrelated at $\rho_{\text{eff}} = 1/|\mathcal{Y}| = 0.1$, the no-coupling fixed point: the tint is present with identical marginal statistics but carries zero label information, so the learned representation cannot exploit it. Shifts re-draw only the indicator, leaving image content untouched: *decorrelate* (s uniform, independent of y; the primary shift, the marginal-controlled cue-breaking analogue of Experiment 1's sign flip) and *anticorrelate* ($s = y$ with probability $1 - \rho$; descriptive only, reported in full). At $\rho = 0.9$ the two shift distributions coincide, a fact the registration turns into a free pipeline-consistency check (Table 2, last row). Model: a 9-block CNN (1,739,210 parameters), 30 epochs, fixed recipe; 10 seeds $\times$ 4 coupling cells $\times$ 2 arms $= 80$ training runs, all seeds committed in advance. Representation Z: the 256-dimensional penultimate features. Readout: multinomial logistic regression on Z ($K_{\text{readout}} = 10 \times 257 = 2570$), fit on one stratified half of the test pool (5,000) and evaluated held-out on the other (5,000); the frozen readout is never evaluated on re-tinted versions of images it was fit on. Per cell, three measurements ($\widehat{\text{CE}}_{\text{ID}}$, the frozen $\widehat{\text{CE}}_Q(r_P)$, and a 5-fold Q-refit $\widehat{\text{CE}}_{Q\text{-refit}}$) yield the two differences the criteria consume:

$$\Delta_{\text{info}} := \widehat{\text{CE}}_{Q\text{-refit}} - \widehat{\text{CE}}_{\text{ID}},$$

$$\Delta_{\text{transfer}} := \widehat{\text{CE}}_Q(r_P) - \widehat{\text{CE}}_{Q\text{-refit}},$$

the first estimating the change in the class's population optimum, $\text{CE}^\star_{\mathcal{A}}(Q) - \text{CE}^\star_{\mathcal{A}}(P)$ (Definition 2.3), the information accessibility lost under shift, and the second the transfer risk of the frozen interface (Definition 2.2): exactly the decomposition the four-quantity vocabulary of Section 2.2 exists to keep separate.

Protocol. The construction pins, training recipe, readout class, seed list, criteria, and thresholds (per-cell tolerance $\delta_{\text{cell}} = \max(2 \times SE_{\text{seed}},\ 0.10 \text{ bits})$, with

Figure 7: Experiment 3 stimuli, built by the pinned construction module (SHA-256 in Appendix C.4) from CIFAR-10 test images (fixed-seed selection, six classes). Top row: clean originals. Middle row: the naive arm's view at $\rho = 0.9$; the additive class-indexed chromatic tint (strength 0.20) is the *entire* manipulation, and its indicator agrees with the true label with probability ρ. Bottom row: the lawful (control) arm's view, which receives the same tint marginal with the indicator drawn independently of the label, so the identical manipulation carries zero label information.

a statistical-power clause) were committed before any naive arm was trained or any shift set built, and the committed analysis script ran exactly once. One validation step (a selftest of the cross-entropy estimator against planted ground truth) was found by its own committed check to validate the wrong estimand and was corrected before any outcome was unblinded, the single dated amendment the registered policy permits (full record, including the preserved failure: Appendix B.5). One consequence carries to every number below: at this readout dimension the absolute held-out cross-entropy is not a validated estimator of the population optimum, while the cross-entropy *differences* the criteria consume (Δ_{info}, Δ_{transfer}, and between-arm gaps) are recovered to the stated tolerance. Absolute values are therefore reported as raw descriptive losses, and pass/fail claims rest only on validated differences.

Results. All five committed criteria pass (Table 2); the verdict rule committed in the registration (all four pass/fail criteria must pass) is met, so the masking signature is *reproduced at scale*. An independent confirmatory re-run — a fresh 80-cell sweep across all four couplings on separate hardware — reproduces the committed seed-means within ϵ (gate-level $\epsilon \leq 0.05$ bits, 0.026 bits on the gated decorrelate cell; per-cell ≤ 0.31 bits), every gate verdict unchanged. Table 3 gives the naive arm's cell means under the primary shift and Figure 8 the masking curve, computed from the committed per-seed results file.

Table 2: Experiment 3: the five pre-registered criteria, committed threshold against measurement (bits, seed-means over all 10 seeds; decorrelate shift). Archive labels in parentheses (Table 8). δ is the committed per-cell formula $\max(2 \times SE_{\text{seed}}, 0.10)$; the 0.10-bit floor bound everywhere except the naive $\rho = 0.99$ transfer cell ($\delta = 0.1177$), far inside the 0.25-bit underpowered trigger, which never fired.

criterion	committed threshold	measured	verdict
E3.1 (`G-B1`) masking-ID premise	naive − lawful $\widehat{\text{CE}}_{\text{ID}} \le +\delta$ at every ρ	−0.1017 / −0.1901 / −0.2295 / −0.3018	**PASS**
E3.2 (`G-B2`) naive transfer brittleness	$\Delta_{\text{transfer}} \ge 0.5$ bits at each $\rho \in \{0.9, 0.95, 0.99\}$	0.5524 / 1.0985 / 3.5603	**PASS**
E3.3 (`G-B3`) lawful arm flat	lawful $\lvert\Delta_{\text{info}}\rvert, \lvert\Delta_{\text{transfer}}\rvert \le \delta$ at every ρ	largest deviation 0.0206, both shifts	**PASS**
E3.4 (`G-B4`) masking-curve monotonicity	Δ_{transfer} non-decreasing in ρ (slack δ)	0.1020 < 0.5524 < 1.0985 < 3.5603, strict	**PASS**
E3.5 (`C-B5`) $\rho = 0.9$ coincidence	dec/anti frozen $\widehat{\text{CE}}_Q(r_P)$ agree within δ (validity check)	0.0014 (naive), 0.0050 (lawful)	consistent

Table 3: Experiment 3, naive arm under the primary decorrelate shift (bits, seed-means over 10 seeds; cross-seed SEs 0.0035–0.0588). $\widehat{\text{CE}}$ columns are raw descriptive held-out losses: at this readout dimension the absolute held-out cross-entropy is not a validated estimator of the population optimum, and pass/fail claims rest only on the validated differences (Section 5.3); $\widehat{\text{CE}}_Q(r_P)$ is the frozen transfer cross-entropy (Definition 2.2). The lawful arm's corresponding $\widehat{\text{CE}}_{\text{ID}}$ is 0.4102–0.4190 across cells with $\lvert\Delta_{\text{info}}\rvert, \lvert\Delta_{\text{transfer}}\rvert \le 0.0206$ everywhere, both shifts.

ρ	$\widehat{\text{CE}}_{\text{ID}}$	$\widehat{\text{CE}}_Q(r_P)$	Δ_{info}	Δ_{transfer}
0.70	0.3154	0.6685	0.2511	0.1020
0.90	0.2266	1.2874	0.5084	0.5524
0.95	0.1807	1.9646	0.6854	1.0985
0.99	0.1172	**4.7737**	1.0962	**3.5603**

Three readings of the table and figure. *First, the masking premise holds in its amplified form.* In-distribution, the naive arm is *better* than the lawful arm at every coupling, by a margin growing to −0.30 bits at $\rho = 0.99$, where the committed construction arithmetic (the cue-only anchor against the realized lawful floor) put the expected gap at ≈ -0.31: the measured value lands within 0.005 bits of the construction's own expectation. In-distribution evaluation rewards the insufficiency rather than exposing it. *Second, the collapse under shift is large, graded, and interface-borne at the top.* The frozen interface's Δ_{transfer} rises strictly (0.1020, 0.5524, 1.0985, 3.5603 bits; 95% CIs $[0.091, 0.115]$, $[0.518, 0.590]$, $[1.036, 1.156]$, $[3.448, 3.669]$), while the lawful arm's worst deviation anywhere, either shift, is 0.0206 bits. At $\rho = 0.99$ the frozen $\widehat{\text{CE}}_Q(r_P)$ reaches 4.7737 bits (5.2748 under the descriptive anticorrelate shift), above the $\log_2 10 = 3.3219$-bit ten-class entropy bound, which is the Section 2.2 tell that this number is a property of the fitted interface (quantity $\text{CE}_Q(r_P)$), not an entropy of anything. As a fraction of the committed cue-only envelope (the closed-form collapse of an idealized calibrated cue reader), the trained system realizes $f(\rho) = 0.1089, 0.1742, 0.4082$ at $\rho = 0.9, 0.95, 0.99$: inside the committed band $[0.1, 1]$ at every point, grazing its lower edge at $\rho = 0.9$. *Third, the validity checks held.* The $\rho = 0.9$ coincidence check agreed at the millibit level (0.0014 and 0.0050 bits); every readout fit converged; and the lawful arm's four identical-data replicate trainings per seed put CNN training nondeterminism at ≈ 0.025 bits mean per-seed range, an order of magnitude inside the smallest threshold any criterion uses, and a free calibration number for future real-data protocols (further checks in Appendix C).

5.3.1 Cue cannibalization: the deviation, and what it bounds

One committed prediction failed, and it is the most informative number in the experiment. The preregistration predicted, descriptively (with no pass/fail threshold attached), $\Delta_{\text{transfer}} \gg \Delta_{\text{info}}$ at $\rho \ge 0.9$, ratio ≥ 2: the collapse should be interface-borne, not information-borne, as in Experiment 1. Measured ratios: 1.09 at $\rho = 0.9$, 1.60 at 0.95, 3.25 at 0.99. The prediction fails at the two moderate couplings and holds only at the extreme one. The mechanism is visible in a pre-registered sanity metric: the naive arm's accuracy on *standard, untinted* CIFAR-10 falls from 0.891 at $\rho = 0.7$ to 0.558 at $\rho = 0.99$ (lawful: 0.924–0.925 everywhere; Figure 9). The naive CNN does not add a cue route on top of intact image features; it *cannibalizes* image-feature learning, so that under shift even a refit readout finds less accessible label information in Z: Δ_{info} reaches 1.0962 bits at $\rho = 0.99$, within its committed bound (the cue's genuine in-distribution information value, 3.2094 bits there) but not small. This deviation *bounds* the paper's interface-borne story, and the text should say so plainly: the direction survives everywhere the prediction applied ($\Delta_{\text{transfer}} > \Delta_{\text{info}}$ at every $\rho \ge 0.9$), but

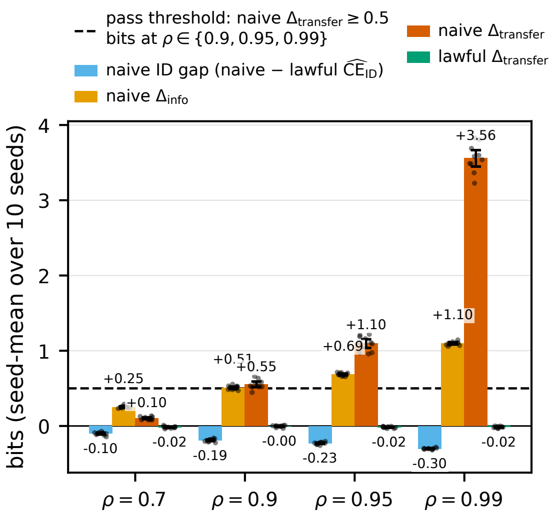


Figure 8: Experiment 3 masking curve, computed from the committed per-seed results file (no measured value is transcribed by hand): for each train-time coupling ρ, the naive arm's in-distribution gap to the lawful arm (negative bars: the insufficient arm is *rewarded* in-distribution), the naive arm's Δ_{info} and Δ_{transfer} under the primary decorrelating shift, and the lawful arm's Δ_{transfer} (flat). Bars are seed-means over the 10 paired seeds; whiskers are 2000-resample bootstrap 95% CIs of the seed-mean; dots are per-seed values. The dashed line is the committed 0.5-bit transfer-brittleness threshold (criterion E3.2, Table 2).

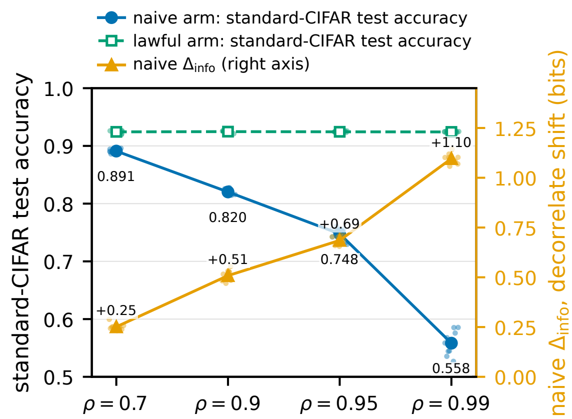


Figure 9: Cue cannibalization, computed from the committed campaign artifacts (per-seed accuracies from the results summary, per-seed Δ_{info} from the results file; nothing transcribed by hand): against train-time coupling ρ, the naive arm's accuracy on *standard, untinted* CIFAR-10 test images (left axis) falls from 0.891 to 0.558 while the lawful arm stays flat at 0.924–0.925 (dashed reference), and the naive arm's Δ_{info} under the primary decorrelating shift (right axis, bits) rises in step. Lines are seed-means over the 10 seeds, dots per-seed values. At high coupling the cue does not ride on top of intact image features; it *displaces* image-feature learning, so part of the shift damage is representation-borne (Section 5.3.1).

the quantitative form "brittleness is interface-borne, not information-borne" is correct only at extreme coupling; at moderate coupling the collapse splits between interface brittleness and genuine representation damage in comparable parts. Experiment 1's clean separation is a property of its frozen *linear* readout over a fixed representation; when the representation itself is learned against the cue, the cue reshapes the representation too. The decomposition that exposes this (Δ_{transfer} against Δ_{info}, kept separate by the Section 2.2 vocabulary) is exactly what an evaluation conflating the four quantities could not have reported.

Scope: one dataset, one architecture, one nuisance design, one readout class, constructed shifts; nothing here is a claim about natural spurious correlations (Section 6.3).

6 Discussion

The recent turn toward observer-dependent information has two complementary edges: how much reusable *structure* a computation-bounded learner can extract from data—measured by epiplexity in concurrent work of Finzi et al. [18]—and how much *task-predictive* information a readout-bounded observer can actually use, which is the edge this paper prices exactly (Section 6.2), together with what deploying that readout costs under distribution shift.

Recommendation: report frozen-versus-refit pairs. The paper's practical output is a reporting practice, not a theorem. For a system whose readout is fit under one distribution and deployed where the distribution may shift, evaluate the relevant shifts twice: once with the deployed readout frozen, once with the readout refit under the shifted distribution, and report the pair (Figure 10). The frozen number is what deployment will experience; the refit number is what the representation still supports; their difference is the price of the readout interface, and it is invisible to any protocol that reports only one of the two. In-distribution evaluation cannot substitute for the pair at any sample size: in both masking experiments it did worse than fail to expose the fragile models; it actively preferred them.

Why the pair is necessary. In-distribution testing estimates one of the four quantities of Section 2.2 (the best cross-entropy a soft readout can attain under the training distribution) and estimates it well: in every validated cell the measurement landed on its analytic

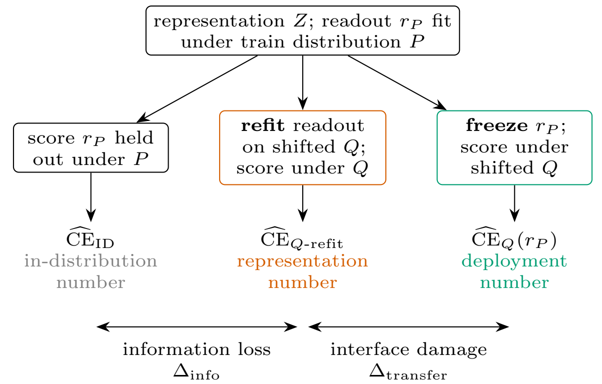


Figure 10: The recommended frozen-versus-refit reporting pair. A readout r_P is fit on the representation under the training distribution P; its held-out score there is the in-distribution number. For a shift of interest Q, score twice: with r_P *frozen*, giving the deployment number $\widehat{\mathrm{CE}}_Q(r_P)$, the loss the deployed interface actually delivers under the shift; and with the readout *refit* under Q on the same representation, giving the representation number $\widehat{\mathrm{CE}}_{Q\text{-refit}}$, the best the representation still supports for that readout class. The pair separates the two failure modes that single-number evaluation conflates: frozen − refit (Δ_{transfer}) is interface damage, the fitted readout no longer matching the shifted data; refit − ID (Δ_{info}) is information loss, a genuine change in the label information the class can access in the representation.

target to within the predicted estimator bias. But what deployment experiences under shift is a different quantity, the transfer cross-entropy of the *frozen* interface, and no in-distribution measurement bounds it. The gap between the two is not a technicality: in the constructed channel, an insufficiency of 0.0004 bits became ≈ 1.7 bits of transfer damage the moment the nuisance–signal coupling moved; on real images, the cue-reliant networks measured better than the control under every standard test and collapsed only under the frozen-readout audit.

Interface damage and representation damage. Experiment 3's decomposition adds a caution that the constructed channel alone would have missed, and it comes from this paper's one failed prediction (Section 5.3.1): at extreme coupling the collapse is carried almost entirely by the fitted interface, as in Experiment 1, where the representation is fixed by construction; at moderate coupling a network trained against a strong cue also learns less image structure, and the interface and representation damage are comparable in size. The two components are separable by measurement (that is what the frozen-versus-refit difference pair does), but both are real.

6.1 External replication on a pretrained language model

The experiments above use a constructed channel and from-scratch CIFAR networks, where ground truth is exact. As a post-hoc external check (not preregistered, unlike Experiments 1–3), we ask whether the frozen-versus-refit decomposition and the two failure modes it separates survive on a real, off-the-shelf pretrained model: **Pythia-410m** ($\approx$ 300B Pile tokens). The same head-only-refit protocol applies, scored in bits per byte so the numbers are comparable across tokenizations.

Weight quantization is information-borne (an irreducible refit floor). Quantizing the weights and measuring frozen versus head-refit cross-entropy on in-distribution text (WikiText-103) reproduces the information-borne signature: 8-bit is free, and below it a floor appears that head-refit cannot remove, with a sharp onset between 4 and 3 bits (Table 4). Refitting the head recovers a meaningful but bounded fraction (30–57%, the interface-borne part); the remainder is irreducible information loss. On this well-trained model the frontier is earlier and steeper than on the from-scratch toy (which paid 0.007 bits irrecoverable at 4-bit): trained weights use their precision, so they tolerate quantization less — the realistic regime.

Distribution shift is interface-borne, growing with OOD distance. The shift leg requires a genuinely out-of-distribution shift: code and Japanese are *in* Pythia's training mixture, so its frozen readout does not collapse on them and head-refit recovers nothing — the interface-borne mechanism is invisible when the "shift" is in-distribution. Adding low-resource, Pile-underrepresented languages exposes it, and the head-refit-recoverable fraction grows monotonically with how far out-of-distribution the shift is (frozen cross-entropy as the OOD-distance proxy): near zero for mild shifts, up to ≈ 0.27 bits/byte for clearly-OOD languages (Table 5; Pearson 0.93, Spearman 0.86, $n = 8$, Latin-script). At the far end the recovered fraction is only $\approx 17\%$ of the gap — the body genuinely under-represents the language — so very-OOD shift is mixed, a real interface-borne component over a representation-borne residual; the frozen-versus-refit pair reads off both.

What the replication adds. Both failure modes the paper isolates on systems with exact ground truth appear on a real pretrained model, separated by the same frozen-versus-refit pair: weight quantization sits on the information-borne side (an irreducible floor), distribution shift on the interface-borne side (refit-recoverable, in proportion to OOD distance),

bits	total	recoverable (interface)	irrecoverable (info)	% interface
8	0.001	—	0.001	free
4	0.389	0.222	**0.166**	57%
3	1.990	0.601	**1.390**	30%
2	2.451	0.894	**1.557**	36%

Table 4: Weight quantization on Pythia-410m (bits/byte, in-distribution; reference is the 16-bit head-refit score). Head-refit recovers the interface-borne part; the irrecoverable column is the information-borne floor, with a sharp 4-to-3-bit onset.

shift	frozen (OOD-ness)	refit recovery
code	0.68	+0.003
Japanese	1.10	+0.021
Vietnamese	1.24	+0.042
Indonesian	1.54	+0.063
Finnish	1.57	+0.031
Yoruba	2.22	**+0.295**
Swahili	2.34	**+0.252**
Welsh	2.46	**+0.274**

Table 5: Distribution shift on Pythia-410m (bits/byte). The head-refit-recoverable (interface-borne) part is near zero for in-distribution/mild shifts and grows to a quarter-bit for genuinely-OOD languages; recovery tracks OOD distance (Pearson 0.93). Same-script-width corpora only (a multi-byte-script point is excluded as a bits-per-byte artifact).

and activation quantization is a distinct mixed locus ($\approx 40\%$ interface-borne). The boundary condition is the honest part: the shift leg is interface-borne only when the shift is genuinely out-of-distribution, which the OOD-distance curve makes quantitative. Caveats: a single model family and scale; round-to-nearest fake quantization; and bits per byte is comparable only across same-script-width corpora.

6.2 Relation to prior work

Classical sufficiency and the DPI. Theorem 3.3 is classical: sufficiency as factorization of $p(Y \mid X)$ through a statistic goes back to Fisher [2] and the Neyman factorization criterion [3], in measure-theoretic form Halmos–Savage [4]; the information-theoretic characterization of sufficiency and the data-processing inequality with its equality condition are textbook material [1, 5, 12]. Theorem 3.4 is the DPI equality condition with the null-set discipline made explicit; the divergence identity of Lemma 3.2 is the standard decomposition $I(Y;X) - I(Y;Z) = I(Y;X \mid Z)$ in conditional-distribution form. Proposition 3.6's separability core is Minsky–Papert [6]; the sine-shattering phenomenon behind Proposition 4.10 is the standard infinite-VC example [11]; sparse-recovery uniqueness under the spark condition is Donoho–Elad [9], with incoherence-based recovery in [10].

Computation in superposition. The Section 4.3 family is a deliberately minimal instance of the setting studied in the superposition-capacity literature; in particular, Borobia, Seguí-Mas, and Tormo-Carbó [19] reconcile linear-readout cross-talk lower bounds (a Welch-type floor) with thresholded Boolean recovery reaching a different capacity regime under sparsity. We cite this line as context for why the family is the right toy, not as support for any claim here; everything we assert about the family is proved exactly (Theorem 4.5) or measured under pre-registration (Section 5.2). Task-aware subspace selection in practice (e.g. Fisher-guided LoRA initialization [20]) is likewise context: it is what lawful-compression questions look like at deep-learning scale, a regime this paper explicitly does not address.

Usable information and probing. The closest formal predecessor is predictive $\mathcal{V}$-information. Xu et al. [14] define observer-constrained entropy and information by optimizing log-loss over a restricted predictive family, thereby making "usable information" depend on the modeling power available to the observer; Ethayarajh et al. [17] carry the same construction to grading dataset difficulty as $\mathcal{V}$-usable information. Our $\mathrm{CE}^{\star}_{\mathcal{A}}$ is the analogous population log-loss optimum for a declared soft readout class, and hard $H_{\mathcal{A}}$ can be viewed as the calibrated-partition specialization induced by deterministic readouts. The contribution here is not the general idea that usable information depends on the observer. It is the exact finite pricing problem: for a declared readout family and declared K/E accounting, what entropy floor remains, and what minimal budget makes that floor fall? We also separate these optimized population quantities from $\mathrm{CE}_Q(r_P)$, the frozen loss of a readout trained under P and deployed under Q, which is not a $\mathcal{V}$-information quantity and is the object exposed by the masking experiments.

A second ancestor is the probing literature. Hewitt and Liang [13] show that probe performance can reflect probe capacity rather than representation content, using control tasks and selectivity; Voita and Titov [15] propose MDL probing to account for the effort needed by a probe; and Pimentel et al. [16] cast probing as information-theoretic estimation. We share the premise that readout complexity matters. The difference is that this paper does not use probe performance as a loose diagnostic of representation content: it treats the readout interface as a declared mathematical object, proves exact interface floors, and measures frozen-versus-refit deployment loss in bits.

Structural information and computational budgets. Concurrent work of Finzi et al. [18] introduces *epiplexity*, the structural half of a time-bounded two-part code: under a fixed compute budget it separates the information in a dataset into the description length of the best time-bounded model and the residual predictive loss, and uses the former to study data selection and out-of-distribution transfer. That decomposition is complementary to ours on two axes. First, within the structure/randomness split epiplexity targets the *structural* component, whereas $H_{\mathcal{A}}$ and $\mathrm{CE}^{\star}_{\mathcal{A}}$ measure the task-predictive component—which that work explicitly groups with $\mathcal{V}$-entropy and computational pseudoentropy as the "random" half it sets aside. Second, epiplexity bounds *computation time* and is task-agnostic, while we bound a declared *readout class* on a fixed task and price the residual entropy floor exactly; and our deployment object $\mathrm{CE}_Q(r_P)$, with its interface-borne-versus-representation-borne split under shift, has no analog in a task-agnostic structural measure. The two compose: epiplexity asks whether data carries reusable structure at all; the price of readout asks how much of it a bounded readout converts to task performance, and whether the shift-induced collapse is fixable at the interface.

What is organization and what is new. Section 3 claims no novelty whatsoever: it is a clean restatement of standard DPI equality/sufficiency results, organized for the accounting question, and a reader who takes it as classical is reading it correctly. Nor do we claim the idea of observer-dependent usable information: $H_{\mathcal{A}}$ and $\mathrm{CE}^{\star}_{\mathcal{A}}$ are deliberately specialized, task-facing descendants of the $\mathcal{V}$-information lineage above, chosen so that exact finite floors and cost frontiers can be proved. What we believe is new is, first, the exact pricing problem and its closures: for a declared readout class and a declared K/E accounting, what entropy floor remains and what minimal budget fells it (Theorems 4.4 and 4.5, Propositions 4.7–4.20, with the open residue isolated in Open Problem 4.22); and, second, the deployment split: $\mathrm{CE}_Q(r_P)$ is not an information quantity at all, and the frozen-versus-refit decomposition built on it exposes failures hidden both by in-distribution testing and by any evaluation that refits the readout under the shifted distribution, quantified under pre-registration from constructed channels up to CNN scale (Sections 5.1 and 5.3). The four-quantity separation of Section 2.2, including the demonstration that lawful compression can strictly degrade the interface-constrained quantity while preserving the information-theoretic one (Remark 3.7), is the operational framework that keeps both claims well-posed.

6.3 Limitations and non-claims

- **Constructed ground truth; constructed coupling at scale.** Experiments 1 and 2 use generative models built so that sufficiency, insufficiency, and geometry-dependence are theorems of the construction. This is by design: it is what makes analytic prediction and exact floors possible. Experiment 3 carries the protocol to real images and learned CNN representations, but its spurious coupling is still constructed (a class-indexed tint at one fixed strength, injected and shifted by a pinned module): the construction supplies the ground truth about the cue that natural data would not. Together the experiments demonstrate that the pre-registered measurement protocol executes end-to-end with admissible proxies from analytic toys up to CNN scale; they establish nothing about natural (non-constructed) spurious correlations.
- **Sufficiency by construction, not inference.** No method for *certifying* sufficiency of a learned representation is proposed or evaluated. The theorems say what sufficiency buys; deciding sufficiency in the wild is a separate (and hard) problem. (Experiment 3's lawful arm is sufficiency-by-construction *for the semantic task*: the tint is made uninformative, not inferred to be.)
- **No production-scale claims.** Experiment 3 trains a 1.74M-parameter CNN on CIFAR-10 (learned features on real images, no longer a constructed channel), but nothing here is evidence about large models or production systems: one dataset, one architecture, one nuisance design, one readout class. The superposition family shares structure with settings studied at scale; the shared structure is the reason for the choice of toy, not a transfer claim.
- **The universal accounting statement is refuted; the narrowed statement is settled; only the residue is open.** Both universal forms of the no-free-improvement claim are *false* (Propositions 4.8 and 4.10, Remark 4.9); the statement narrowed to affine-threshold compo-

sitions is settled at the linear budget (Proposition 4.12), pinned at $E = 12$ for the XOR family (Proposition 4.14), and closed completely for the superposition family (Propositions 4.19 and 4.20); what remains open is the named residue (Open Problem 4.22). The proved payment floors (Proposition 4.7) are family- and convention-restricted: the structure is convention-robust, the constants are not. Universal-accounting language should not be read into any result here.

- **The refutation conditions remain in force.** The events that would refute the claims here are stated precisely and remain testable; each survived one test (two, counting Experiment 3's scaled rerun of the masking conditions). Their proved theorem-side referents, and what an apparent future occurrence would localize to, are recorded in Appendix B.3. Survival is cumulative, not a closed book.

7 Reproducibility

All reproduction materials (the paper source and figures, all three pre-registrations, the experiment scripts, the raw per-seed results, and the verification scripts) are publicly available in a version-controlled reproduction repository [21] at the annotated tag `v1.0-submission`. The before-any-run ordering of each experiment's evidence chain (pre-registration, then the once-only analysis script, then results) is externally timestamped via OpenTimestamps (Bitcoin-anchored, hash-only): the commit ancestry below records the ordering, and timestamped evidence manifests, whose attestation files ship in the repository's `evidence/` directory, additionally pin it — Experiments 1 and 2 dated 2026-06-10, Experiment 3 dated 2026-06-11/12, plus a combined paper manifest dated 2026-06-15. A reviewer can verify the entire chain today. The paper is self-contained: every theorem is stated and proved in full in the body or appendices, and every measured number traces to the committed artifacts below. The repository stores the experiment materials under paper-native directory and file names; Table 8 (Appendix B) gives the complete map, for reproduction only.

For each experiment the pre-registration (construction, fixed parameters, seed streams, proxy conventions, analytic predictions, pass/fail thresholds, amendment policy) was committed before any experimental sample was drawn, and the results were committed with raw per-seed data. Full commit hashes:

- Experiment 1 (lawful compression), archive directory `mlr-proof-program/experiments/track3_gate2/`: pre-registration commit `3b05461418dc562212588562acb0aa9a18191c1b`, results commit `341851624b3b0049d2f116524b7403a9fca5bb6d` (script `lawful_compression_experiment.py`; raw data `results.csv`, 110 rows).
- Experiment 2 (readout ladder), archive directory `mlr-proof-program/experiments/track4_gateB/`: pre-registration commit `6a3645ebc81b04b25b726276e95469b3afc8d8d6`, results commit `1dca8225c7e4a6553b58b6b53d7e691eec639ec4` (script `readout_geometry_experiment.py`; raw data `results.csv`, 120 rows).
- Experiment 3 (masking at scale), archive directory `mlr-proof-program/experiments/expB_masking_real/`: pre-registration commit `d26850e3a86ce1b02aa1f5d1f474489369f5d669`, pre-unblinding amendment commit `dc7b1fd4d2d5c14aa0aa4c558a49f5f1122e58f6`, results commit `23a90152676e6c98b2a179a6d7e2ed9904c2e714`, raw-artifact commit `478df16883aca05dea0bf5b28f852494f0f66c67` (construction module `spurious_cifar.py`, SHA-256 pinned in Appendix C; training script `train_run.py`; once-only analysis script `analyze_masking.py`, committed pre-run; raw data `results.csv`, 800 rows).

The Experiment 1–2 hashes are exactly those pinned by the externally timestamped evidence manifest dated 2026-06-10; the Experiment 3 hashes are pinned by the manifest dated 2026-06-11/12. The Experiment 1–2 attestation additionally pins the digest of a full-history bundle of the evidence archive, fixing the entire ancestry of its commits (content, order, and dates) as of the attestation date. All attestation files are public in the repository's `evidence/` directory; a reader can check these claims independently today.

Exact commands. From the repository root, for Experiment 1:

```
cd mlr-proof-program/experiments/track3_gate2
python3 lawful_compression_experiment.py --analytic
```
(committed analytic predictions; Gauss–Hermite quadrature)
```
python3 lawful_compression_experiment.py
```
(full pre-registered run; writes `results.csv`*)*

For Experiment 2 and the machine-checked enumerations:

```
cd mlr-proof-program/experiments/track4_gateB
```

```
python3 readout_geometry_experiment.py --analytic
```
(exact per-seed population floors; convex optimization)

```
python3 readout_geometry_experiment.py
```
(full pre-registered run; writes `results.csv`*)*

```
python3 verify_xor_min_e.py
```
(all machine-checked enumerations of Appendix A; asserts every claim)

```
python3 ../kef_frontier/superposition_min_e.py
```
(the superposition-frontier enumerations and case analyses of Appendix A.4; 23 hard-asserted checks)

For Experiment 3 (the 80 training runs need a GPU; the once-only analysis is CPU-bound):

```
cd mlr-proof-program/experiments/expB_masking_real
python3 train_run.py --arm naive|lawful
  --rho 0.7|0.9|0.95|0.99 --seed 0..9
```
(one of the 80 committed training cells; writes the feature archives)

```
python3 analyze_masking.py --run
```
(estimator selftest, all measurements, and pass/fail evaluation; ran exactly once on the committed protocol; writes `results.csv` *and the summaries)*

One committed script, `make_figures.py` in the paper's `figures/` subdirectory, generates the data-driven figures (Figures 5, 6, 8, 7, and 9): it reads the three `results.csv` files above and the Experiment 3 `results_summary.json` by relative path, builds the stimulus strip by importing the pinned construction module (its SHA-256, Appendix C.4, is asserted before import) against the CIFAR-10 test batch, and writes the figure PDFs under `figures/`; no measured value is transcribed by hand into the script. Regenerate with

`python3 figures/make_figures.py`.

The paper then builds with two passes of `pdflatex -interaction=nonstopmode main.tex` in the directory

`mlr-proof-program/papers/`
`lawful-compression-readout-geometry/`.

Environment. The committed Experiment 1–2 runs used Python 3.12.3 with NumPy 2.4.6 (the only third-party dependency); the verification enumerations of Propositions 4.8–4.10, 4.12–4.14, and Appendix A are implemented in the committed script `verify_xor_min_e.py` (Table 8), which uses only the Python standard library and re-asserts every enumeration claim in under a second; its SHA-256 digest,

`aa9f4eed57e7ccd7acc7936fc2c7dce0`
`6063ea32fee1795843ac3e4d012fd165`,

is pinned, together with the digests of both Experiment 1–2 pre-registration files, by the timestamped evidence manifest above. The superposition-frontier enumerations and case analyses (Lemma 4.16–Proposition 4.20, Appendix A.4) are implemented in the committed script `superposition_min_e.py` (Table 8), likewise standard-library-only (exact `Fraction` arithmetic; 23 hard-asserted checks, seconds of runtime); its SHA-256 digest is

`c8de5f8c4eb3c9573086c14682a0b094`
`256c7c7f2a2f555624467cba7b89b61a`.

Experiment 3 trained with PyTorch 2.12.0 (CUDA 13.0) and ran its readout measurements with scikit-learn 1.9.0 (settings serialized verbatim into every result row). The figure-generation script additionally uses Matplotlib 3.10.9 (Agg backend, vector PDF output); the paper was built with pdfTeX 1.40.25 (TeX Live 2023). Hardware: a 20-core Arm (Cortex-X925) Linux workstation with 121 GiB RAM; no GPU is used for Experiments 1–2 or any enumeration. Experiment 3's 80 training runs used two NVIDIA GB10 (Arm) nodes, one GPU per run; its readout measurements are CPU-only.

All randomness flows through declared `numpy.random.default_rng` streams, with seeds and stream offsets fixed in the pre-registrations, so every number in Tables 1, 9, 10, and 11 is deterministically reproducible by re-running the committed scripts. Experiment 3's construction and shift draws are likewise exactly reproducible from their declared streams, but GPU training is not bit-deterministic; the committed lawful-arm replicate design measures this nondeterminism directly, at ≈ 0.025 bits mean per-seed range, an order of magnitude inside the materiality floor (Section 5.3). Each experiment was run exactly once; Experiments 1 and 2 with no amendments, Experiment 3 with the single dated pre-unblinding amendment its registered policy permits (Appendix B).

References

[1] T. M. Cover and J. A. Thomas. *Elements of Information Theory*. Wiley, 2nd edition, 2006.

[2] R. A. Fisher. On the mathematical foundations of theoretical statistics. *Philosophical Transactions of the Royal Society A*, 222:309–368, 1922.

[3] J. Neyman. Su un teorema concernente le cosiddette statistiche sufficienti. *Giornale dell'Istituto Italiano degli Attuari*, 6:320–334, 1935.

[4] P. R. Halmos and L. J. Savage. Application of the Radon–Nikodym theorem to the theory of sufficient statistics. *Annals of Mathematical Statistics*, 20(2):225–241, 1949.

[5] S. Kullback. *Information Theory and Statistics*. Wiley, 1959.

[6] M. Minsky and S. Papert. *Perceptrons*. MIT Press, 1969.

[7] S. Muroga. *Threshold Logic and its Applications*. Wiley, 1971.

[8] O. Kallenberg. *Foundations of Modern Probability*. Springer, 2nd edition, 2002.

[9] D. L. Donoho and M. Elad. Optimally sparse representation in general (nonorthogonal) dictionaries via ℓ^1 minimization. *Proceedings of the National Academy of Sciences*, 100(5):2197–2202, 2003.

[10] J. A. Tropp. Greed is good: algorithmic results for sparse approximation. *IEEE Transactions on Information Theory*, 50(10):2231–2242, 2004.

[11] V. N. Vapnik. *Statistical Learning Theory*. Wiley, 1998.

[12] Y. Polyanskiy and Y. Wu. *Information Theory: From Coding to Learning*. Cambridge University Press, 2024.

[13] J. Hewitt and P. Liang. Designing and interpreting probes with control tasks. In *Proceedings of EMNLP-IJCNLP*, 2019. arXiv:1909.03368.

[14] Y. Xu, S. Zhao, J. Song, R. Stewart, and S. Ermon. A theory of usable information under computational constraints. In *International Conference on Learning Representations (ICLR)*, 2020. arXiv:2002.10689.

[15] E. Voita and I. Titov. Information-theoretic probing with minimum description length. In *Proceedings of EMNLP*, 2020. arXiv:2003.12298.

[16] T. Pimentel, J. Valvoda, R. Hall Maudslay, R. Zmigrod, A. Williams, and R. Cotterell. Information-theoretic probing for linguistic structure. In *Proceedings of ACL*, 2020. arXiv:2004.03061.

[17] K. Ethayarajh, Y. Choi, and S. Swayamdipta. Understanding dataset difficulty with $\mathcal{V}$-usable information. In *Proceedings of ICML*, 2022. arXiv:2110.08420.

[18] M. Finzi, S. Qiu, Y. Jiang, P. Izmailov, J. Z. Kolter, and A. G. Wilson. From entropy to epiplexity: rethinking information for computationally bounded intelligence. arXiv preprint arXiv:2601.03220, 2026. https://arxiv.org/abs/2601.03220.

[19] H. Borobia, E. Seguí-Mas, and G. Tormo-Carbó. Linear-readout floors and threshold recovery in computation in superposition. arXiv preprint arXiv:2605.01192, 2026. https://arxiv.org/abs/2605.01192.

[20] Z.-Q. Feng, Y.-J. Lin, and H.-Y. Kao. Learning in the Fisher subspace: a guided initialization for LoRA fine-tuning. arXiv preprint arXiv:2605.01046, 2026. https://arxiv.org/abs/2605.01046.

[21] CURV Institute. The Price of Readout — reproduction repository: pre-registrations, experiment artifacts, raw per-seed data, verification scripts, paper sources, and the externally timestamped evidence manifests. Publicly available, 2026; annotated tag `v1.0-submission`. https://github.com/curv-institute/price-of-readout.

A Finite Enumerations: Exact Floors and the Minimal-E Edge

Both exact floors rest on finite, exhaustively checkable enumerations. This appendix tabulates them in full so that the proofs of Proposition 3.6 and Theorem 4.5(ii) can be verified by inspection, and documents the exhaustive circuit enumeration behind Proposition 4.14 (Section A.3) and the breakpoint and skeleton enumerations behind the superposition frontier (Section A.4). Every machine check claimed in this appendix, and every other machine check claimed in the paper, is re-run with hard assertions by two committed scripts in the reproduction archive (Section 7): `experiments/track4_gateB/verify_xor_min_e.py` for Propositions 4.8, 4.10, 4.12, 4.14, Remark 4.9, and Theorem 4.5(iii); and `experiments/kef_frontier/superposition_min_e.py` for Lemma 4.16, Propositions 4.18–4.20, and Remark 4.21.

A.1 Halfspace partitions of the four Boolean points (Proposition 3.6)

The four points of $\{0,1\}^2$ carry mass 1/4 each and labels $Y = z_1 \oplus z_2$. A readout $r \in \mathcal{A}_{\text{lin}}(2)$ induces the partition $\{B, B^c\}$ with $B = \{z : w \cdot z \geq \theta\}$; B and B^c give the same partition. Table 6 lists *every* partition of the four points into two cells of the form (subset, complement), whether it is halfspace-achievable, a realizing (w, θ) where one exists, and the resulting conditional entropy. There are $2^4/2 = 8$ subset/complement partitions (16 subsets, each partition counted by an unordered complementary pair): the trivial partition (1), the singleton/triple partitions (4), and the three 2-2 splits, namely the two coordinate splits and the diagonal split (3). By the symmetries of the construction they fall into the five rows shown; the row counts are in partitions, with the corresponding subset counts alongside, and they sum to 8 partitions (16 subsets).

Table 6: Exhaustive halfspace-partition enumeration for the XOR floor. Labels are listed in the point order $(0,0), (0,1), (1,0), (1,1)$, i.e. $Y = (0,1,1,0)$. Counts are in partitions (unordered complementary pairs of subsets), with the corresponding subset counts alongside; in particular the two diagonal-pair subsets are complements of one another and form a *single* partition. The achievable minimum is $\frac{3}{4}h_2(\frac{1}{3}) \approx 0.6887$ bits; the only partition that would beat it is the diagonal one, and it is exactly the non-achievable one (Proposition 3.6, Step 1).

partition $\{B, B^c\}$	partitions	subsets	achievable	example (w, θ)	$H(Y \mid r(Z))$
$\emptyset$ vs all four	1	2	yes	$w = (0,0)$, $\theta = 1$	$H(Y) = 1$
singleton vs triple	4	8	yes	$B = \{(1,1)\}$: $w = (1,1)$, $\theta = \frac{3}{2}$	$\frac{3}{4}h_2(\frac{1}{3}) \approx 0.6887$
adjacent pair (z_1 split)	1	2	yes	$B = \{z_1 = 1\}$: $w = (1,0)$, $\theta = \frac{1}{2}$	$\frac{1}{2} \cdot 1 + \frac{1}{2} \cdot 1 = 1$
adjacent pair (z_2 split)	1	2	yes	$B = \{z_2 = 1\}$: $w = (0,1)$, $\theta = \frac{1}{2}$	1
diagonal pair vs diagonal pair	1	2	**no**	— (Step 1 of Prop. 3.6)	(0; excluded)
total	8	16			

Each singleton cell is Y-pure and its complementary triple carries label proportions 2:1, giving the common value $\frac{3}{4}h_2(\frac{1}{3})$ for all four singleton/triple partitions. The infimum over $\mathcal{A}_{\text{lin}}(2)$ is the minimum over the achievable rows: $\frac{3}{4}h_2(\frac{1}{3})$, attained. The corresponding 14 achievable subsets B ($\emptyset$, full set, 4 singletons, 4 triples, 4 adjacent pairs, realizing the 7 achievable partitions) were confirmed by machine enumeration over (w, θ); the two diagonal-pair subsets never occur.

A.2 Prefix/suffix splits of the six superposition atoms (Theorem 4.5(ii))

The six atoms $z = 3, 5, 6, 9, 10, 12$ carry mass 1/6 each and labels $(1,1,0,1,0,0)$ in sorted order. A readout $r \in \mathcal{A}_{\text{lin}}(1)$ realizes exactly the prefix/suffix partitions: prefix cell $\{z_{(1)}, \ldots, z_{(m)}\}$ versus the rest, $m = 0, \ldots, 6$ (complements coincide). Table 7 computes $H(Y \mid r(Z))$ for every m.

Table 7: Exhaustive prefix/suffix enumeration for the superposition floor. "prefix labels" / "suffix labels" list the Y values in each cell; the entropy is $\frac{m}{6}h_2(p_{\mathrm{pre}}) + \frac{6-m}{6}h_2(p_{\mathrm{suf}})$ with p the cell's label-1 fraction. The minimum, $\frac{2}{3}h_2(\frac{1}{4}) \approx 0.5409$ bits, is attained at $m = 2$ and $m = 4$.

m	prefix labels	suffix labels	expression	value (bits)
0	—	$1,1,0,1,0,0$	$h_2(\frac{1}{2})$	1.0000
1	1	$1,0,1,0,0$	$\frac{1}{6}\cdot 0 + \frac{5}{6}h_2(\frac{2}{5})$	0.8091
2	$1,1$	$0,1,0,0$	$\frac{2}{6}\cdot 0 + \frac{4}{6}h_2(\frac{1}{4})$	**0.5409**
3	$1,1,0$	$1,0,0$	$\frac{1}{2}h_2(\frac{1}{3}) + \frac{1}{2}h_2(\frac{1}{3})$	0.9183
4	$1,1,0,1$	$0,0$	$\frac{4}{6}h_2(\frac{3}{4}) + \frac{2}{6}\cdot 0$	**0.5409**
5	$1,1,0,1,0$	0	$\frac{5}{6}h_2(\frac{3}{5}) + \frac{1}{6}\cdot 0$	0.8091
6	$1,1,0,1,0,0$	—	$h_2(\frac{1}{2})$	1.0000

Since $\mathcal{A}_{\mathrm{lin}}(1)$ realizes only these seven partitions, the infimum equals the tabulated minimum $\frac{2}{3}h_2(\frac{1}{4})$. (For Theorem 4.5(iii), the analogous exhaustive object is the set of $\binom{6}{2} + 6 = 21$ contiguous blocks; their entropies take only the four values 0.5409, 0.8091, 0.9183, 1, as listed in the proof, and the interval/complement structure is also what makes the quadratic-threshold tie of Remark 4.9 an enumeration over the same 21 blocks.)

A.3 The exhaustive circuit enumeration behind Proposition 4.14

Proposition 4.14's minimality half asserts that no affine-threshold composition (Definition 4.11) with $E \leq 11$ computes XOR on $\{0,1\}^2$. The search space is finite; this subsection records its structure so the enumeration is auditable, and the committed script `verify_xor_min_e.py` (under `experiments/track4_gateB/`) executes it from scratch with hard assertions (runtime under one second; Python standard library only).

Threshold-function censuses, validated. On a Boolean domain every affine-threshold gate computes a linear-threshold function (LTF) of its inputs, so gates can be enumerated by truth table rather than by real parameters. The script generates the LTF census for fan-in $f = 0,1,2,3$ by exhausting integer weights $|w_i| \leq 4$ and half-integer thresholds, and validates the resulting counts 2, 4, 14, 104 against the known census of threshold Boolean functions [7] (OEIS A000609, the sequence used by the script). The bounded integer-weight sweep is complete for $n \leq 3$ variables because every threshold function of few variables is realizable with small integer weights (Muroga [7] gives explicit bounds, magnitude ≤ 2 sufficing for ≤ 3 variables), and the census match 2/4/14/104 is the verification that the sweep misses nothing. These are the counts quoted in the proof of Proposition 4.14.

Reductions, machine-checked. Three closure facts prune the search, and each is itself verified by table lookup rather than assumed: (i) the census at each fan-in is closed under complementation (at equal cost, so it suffices to look for XOR and its complement directly); (ii) *repeated-input absorption*: a gate reading a repeated signal computes an LTF of its distinct signals, realizable by a strictly cheaper gate of smaller fan-in; (iii) *constant-input absorption*: a gate reading a constant computes an LTF of its remaining signals. Hence the main enumeration may restrict to gates with distinct, non-constant inputs. Every such gate costs $2f + 1 \geq 3$ FLOPs, so at $E \leq 11$ at most three gates occur; gate k (in topological order) sees $2 + (k-1)$ distinct signals, so fan-in 4 would require a third gate and total cost $\geq 3 + 3 + 9 = 15 > 11$. Fan-ins are therefore at most 3, and the admissible gate fan-in multisets are exactly:

fan-in multiset	$E = \sum_g (2f_g + 1)$	realizable wiring
$\{1\}$	3	yes
$\{2\}$	5	yes
$\{1,1\}$	6	yes
$\{3\}$	7	no (a first gate sees only z_1, z_2)
$\{1,2\}$	8	yes
$\{1,1,1\}$	9	yes
$\{1,3\}$	10	yes (the fan-in-3 gate second)
$\{2,2\}$	10	yes
$\{1,1,2\}$	11	yes

All other multisets cost ≥ 12 or need unavailable signals.

Enumeration and conventions. For each ordered gate sequence consistent with a multiset above, the script enumerates every wiring (unordered selections of distinct available signals; input permutations are absorbed because the census ranges over *all* LTFs of the selected inputs) and every census function per gate, and evaluates the final gate's truth table on the four points. The output convention is that the last gate is the readout; this is exhaustive because every prefix of an enumerated circuit is itself enumerated, so a circuit whose output were an earlier gate is covered at lower cost. The result: the set of Boolean functions achievable at $E \leq 11$ is exactly the 14 halfspace-achievable functions of Table 6 (the same set as at $E \leq 5$, which is the machine check quoted in Proposition 4.12), and in particular neither XOR $(0,1,1,0)$ nor its complement occurs. The script prints the achieved function set at each $E \leq 11$, verifies equality with the 14 halfspace functions of Table 6, and asserts absence of XOR and its complement. The script then verifies the $E = 12$ witness (exact XOR; $5+7$ FLOPs; $K_{\text{unsh}} = 7$, $K_{\text{val}} = 3$), and re-runs the Appendix A.1 and A.2 censuses, the 21-block and cut-pair enumerations of Theorem 4.5(iii), the quadratic-decoder checks of Proposition 4.8 and Remark 4.9, and the sine-decoder margins of Proposition 4.10.

A.4 The breakpoint and skeleton enumerations behind the superposition frontier (Propositions 4.18–4.20)

The superposition frontier rests on three finite structures, each exhaustively enumerated by the committed script `superposition_min_e.py` (under `experiments/kef_frontier/`; Python standard library only, exact `Fraction` arithmetic, 23 hard-asserted checks).

Exact entropies of all 32 partitions. Each conditional entropy of a two-cell partition of the six uniform atoms is a rational combination of 1, $\log_2 3$, and $\log_2 5$, which are $\mathbb{Q}$-linearly independent; the script stores each entropy as its exact coefficient vector, so the five-value census of Proposition 4.18(i) and the uniqueness of the sub-floor partition are decided by exact comparison, not floating point. The per-budget table:

budget	best achievable H	attained by
$b \leq 1$	$\frac{2}{3}h_2(\frac{1}{4}) \approx 0.5409$	prefix split $\{3,5\}$ (Table 7)
$b \leq 2$	$\frac{2}{3}h_2(\frac{1}{4})$	same (best new b=2 pattern: interior singleton, 0.8091)
$b \leq 3$	0	exact decode, pattern 110100

Reduced skeletons with $E \leq 16$. A skeleton fixes the gate count and wiring (which gates read z, which read which earlier gates), not the parameters. With the reduction rules of Proposition 4.19's proof (no fan-in-0 gates, no zero-weight inputs, no unread non-output gates), every reduced gate costs at least 3 FLOPs, so the skeletons with $E \leq 16$ are finite: the script enumerates all of them up to isomorphism (49) in cost order, with each skeleton's unshared count K_{unsh} and the Lemma 4.16 bound on its output breakpoints. The decision-relevant rows:

E	skeleton(s)	max B	verdict
3	$g_1 \leftarrow z$	1	floor binds ($\mathcal{A}_{\text{lin}}(1)$)
6	$g_1 \leftarrow z$; $g_2 \leftarrow g_1$	1	floor binds
7	none (no reduced skeleton has cost 7)	—	floor binds
8	$g_1 \leftarrow z$; $r \leftarrow \{z, g_1\}$, unique	3	witness exists

The script asserts that every skeleton with $E \leq 7$ has output breakpoint bound ≤ 1 (so by Proposition 4.18 its best achievable value is the floor itself) and that the minimum cost of any skeleton with bound ≥ 3 is $E = 8$, with the witness shape unique there. (Randomized sampling over skeleton parameters, 4,000 draws per $E \leq 7$ skeleton, never produced two or more output switches: supporting evidence only; the proof is the bound.)

Witnesses and the distinct-value (K_{val}) case analysis. The two $E = 8$ witnesses of Proposition 4.19 and the paper's $E = 16$ width-3 net (Theorem 4.5(iv)) are each verified at all six atoms in exact rational arithmetic. For the K_{val} bound, the script checks that the split $\{3,5,6\}$ vs. $\{9,10,12\}$ is the unique one-threshold split compatible with the required pattern, then verifies that in all four sign cases every merge pattern reaching $K_{\text{val}} \leq 3$ reduces to one of ten interval conditions, each empty over the admissible ranges (boundary contacts included), and that no $K_{\text{val}} = 3$ witness at $E = 8$ appears in 200,000 random draws (evidence only; the case analysis is the proof). The script also re-runs the width-2 cut-pair enumeration of Theorem 4.5(iii) for continuity with Appendix A.2.

B Pre-Registration Protocol, Criteria, and Amendments

This appendix records the measurement discipline in full: the shared protocol, the per-experiment criterion register with the archive's short labels, the refutation conditions, one criterion downgraded by its own statistical-power provision, the single protocol amendment, and the artifact map. Nothing here is needed to follow the results; it is the audit trail.

B.1 The shared protocol

All three experiments followed the same discipline, and each was executed exactly once (Experiments 1 and 2 with no amendments, Experiment 3 with the single dated, pre-unblinding amendment its registered policy permits; Appendix B.5):

1. **Pre-registration before any run.** The generative construction, fixed parameters, seeds, readout classes, training protocol, measurement conventions for $H/K/E$, analytic predictions, pass/fail thresholds with tolerances, refutation events, and an amendment policy were committed before any experimental sample was drawn to a repository that is version-controlled, externally timestamped, and publicly available (commit hashes in Section 7).
2. **Analytic predictions, not narratives.** The predictions are exact or quadrature-grade computations from the generative model (Gauss–Hermite quadrature for Experiment 1; exact enumeration over a finite atom distribution plus population-level convex optimization for Experiment 2), committed before the run. The experiments test whether trained readouts on sampled data land on those committed numbers out-of-sample. (Experiment 3, whose learned CNN features admit no analytic ground truth, commits instead closed-form *cue-only* anchors from its generative construction plus an estimator-level selftest; Appendix B.5.)
3. **Declared estimation target.** Per Remark 3.7 and Section 2.2, each experiment declares the population quantity its measured cross-entropy estimates: $\mathrm{CE}^{\star}_{\mathcal{A}}$ for a declared soft class (Definition 2.3), together with where that optimum provably equals $H(Y \mid Z)$. Where no estimate is validated (post-shift), the measurement is reported as the transfer cross-entropy $\mathrm{CE}_Q(r_P)$ that it is.
4. **Measurement-quality criteria (P1–P5).** Five properties were required of every measured quantity, committed once for all three experiments. P1: observable operational meaning (bits of held-out cross-entropy; counted parameters; counted FLOPs by stated formulas; wall-clock reported direction-only). P2: coefficient of variation $> 20\%$ across test conditions, so that "no change" findings are measured against real variation. P3: the cross-entropy estimator is validated against analytic ground truth. P4: identical units, readout classes, and protocols across compared arms. P5: no trade-off multiplier used or tuned anywhere; accounting consequences reported in sign-robust form only. All five passed in all three experiments (Appendix C).
5. **Anti-pattern audit.** Results were checked against four pre-stated anti-patterns: in-sample fits; symbol reuse without derivation; tautological identities (no measured quantity is an algebraic function of another); selective reporting (all seeds, cells, and pre-declared descriptive observables reported).
6. **Honest verdicts.** Criteria are graded PASS / FAIL / INCONCLUSIVE against the committed thresholds; the statistical-power provision (Appendix B.4) can downgrade an apparent pass.

B.2 Criterion register

The criteria are labeled E1.1–E1.5, E2.1–E2.5, and E3.1–E3.5 in this paper; Table 8 maps each label to its archived counterpart. In words:

- **E1.1** (task-uncertainty proxy preservation): the seed-mean held-out cross-entropy of the compressed arm S matches the full arm X within $\delta = 0.010$ bits at every d_n. **E1.2** (counted savings): K and E decrease by the constructed factors, with measured training FLOPs equal to the committed formula values. **E1.3** (estimator validation): every (arm, d_n) cell's measured value lands within tolerance of its committed analytic value. **E1.4** (lawful invariance): the lawful arm moves at most δ under the shift. **E1.5** (predicted degradation): the insufficient arm's post-shift transfer cross-entropy lands on the committed analytic prediction. All five passed (Section 5.1).

- **E2.1** (floor reproduction): the trained linear readout reproduces its committed per-seed exact $\mathrm{CE}^\star_{\mathrm{lin}}$ floor within 0.010 bits at every load. **E2.2** (paid improvement): the oracle rung beats the exact linear floors at the declared gap cells, at its counted codebook cost. **E2.3** (trained-network improvement, secondary): see Appendix B.4. **E2.4** (ladder accounting): K and E strictly increase up-ladder at every cell. **E2.5** (refutation non-occurrence): no readout at linear K/E measures below the linear class's exact floor by more than 0.010 bits at any cell or seed. E2.1, E2.2, E2.4, E2.5 passed; E2.3 is inconclusive by its own provision (Section 5.2, Appendix B.4).
- **E3.1–E3.5**: stated in words, with thresholds and measured values, in Table 2. All passed.

B.3 Refutation conditions and their referents

The committed refutation events have proved theorem-side referents. The event "a geometry-ignorant readout matching a geometry-aware readout at equivalent K/E" is governed at the linear budget by Proposition 4.12: an apparent observation of a readout in $\mathcal{A}_{\mathrm{comp}}$ at the linear budget beating a floor on these families would localize to a cost-hiding K/E convention or to a measurement not estimating the stated task, never to an open claim. Above the linear budget such an observation is not a refutation at all: it is a data point on the payment frontier, checkable against the closed characterizations where they exist (Propositions 4.14 and 4.20). An arithmetic-basis readout outside $\mathcal{A}_{\mathrm{comp}}$ beating a floor is an instance of the already-recorded refutation (Proposition 4.8 is itself such an event, supplied by analysis rather than experiment). Experiment 2's criterion E2.5 is therefore an audit of the measurement pipeline (consistency with a proved statement) rather than evidence for an open conjecture (the floor it guards being the soft optimum $\mathrm{CE}^\star_{\mathrm{lin}}$).

B.4 The E2.3 downgrade: a worked example of the discipline

Criterion E2.3 (secondary) asked whether the *trained* width-64 network beats the exact $\mathrm{CE}^\star_{\mathrm{lin}}$ floor by pre-set margins at $d_z \in \{4, 6\}$ (thresholds: seed-mean cross-entropy ≤ 0.4540 and ≤ 0.3599 bits). The point estimates passed at both cells (0.3729 and 0.0569 bits), and every single seed's trained network landed below *that seed's own* exact floor by at least 0.05 bits (20 of 20 seeds; worst margins 0.1096 and 0.2836 bits). Nevertheless E2.3 is reported **INCONCLUSIVE**: the pre-registration's statistical-power provision binds, because the cross-seed seed-mean standard errors (0.0404 and 0.0150 bits) exceed the committed 0.005-bit bound. The provision was written with measurement noise in mind and did not anticipate that real cross-projection heterogeneity (each seed has a different projection, hence a different true floor) would dominate the unpaired variance while paired per-seed margins are decisive. Per the no-massaging rule it is applied as written, against the experimenters' reading of the evidence. Nothing rests on E2.3 (the class-level gap is established by E2.2 and, theorem-side, by Theorem 4.5), but the episode is the protocol working as intended, and it records a concrete lesson for future registrations: pre-register *paired* criteria when per-condition ground truth varies by construction.

B.5 The Experiment 3 amendment

Experiment 3's committed validation selftest required the cross-entropy estimator to recover the *absolute* $\mathrm{CE}^\star$ of planted softmax-linear posteriors within 0.05 bits at the experiment's own $(n, D, |\mathcal{Y}|)$. The selftest **failed** when the committed script first ran, exactly as the registration's own escape clause anticipated it might: the finite-sample excess of the 10×257-parameter multinomial-logistic maximum-likelihood estimator at $n = 5{,}000$ is far above the tolerance (the asymptotic heuristic $K_{\mathrm{readout}}/(2n \ln 2) \approx 0.3708$ bits, Appendix C.1, anticipates the order of magnitude), and the failure was not a script defect. The registered amendment policy (one amendment, dated, pre-unblinding, measurement-quality reasons only, never loosening a pass/fail threshold) was then spent on the fix: the criterion was amended to recovery of planted $\mathrm{CE}^\star$ *differences* (the estimand every pass/fail comparison actually consumes), and the amended selftest passed (errors 0.0056 and 0.0007 bits on a ~ 0.5-bit pair and a null pair). The original FAIL record is preserved verbatim in the committed artifacts. The consequence for reporting is stated in Section 5.3: cross-entropy *differences* are validated estimates; absolute held-out values are reported as raw descriptive losses; the frozen $\widehat{\mathrm{CE}}_Q(r_P)$ is definitionally the deployed-interface loss, though its reported value remains a finite-sample estimate of its population value.

The selftest caught its own design error (validating a more demanding cousin of the estimand the comparisons consume) before unblinding; the lesson (validate the estimand the comparisons consume) is recorded.

B.6 Artifact map

Table 8: Artifact map: names used in this paper against the corresponding names in the reproduction archive (locations relative to `mlr-proof-program/`). The archived pre-registration and results files use the internal labels shown in the middle column; the map is provided for reproduction only.

This paper	Archive label	Archive location
Experiment 1 (lawful compression)	"track 3 / gate 2"	`experiments/track3_gate2/`
Experiment 2 (readout ladder)	"track 4 / gate B"	`experiments/track4_gateB/`
Experiment 3 (masking at scale)	"experiment B" (`expB-masking-real-v1`)	`experiments/expB_masking_real/`
Criteria E1.1 / E1.2 / E1.3	`G2-a` / `G2-b` / `G2-c`	Experiment 1 pre-registration
Criteria E1.4 / E1.5	`G3-a` / `G3-b`	Experiment 1 pre-registration
Criteria E2.1–E2.5	`B1`–`B5`	Experiment 2 pre-registration
Criteria E3.1–E3.4	`G-B1`–`G-B4`	Experiment 3 pre-registration
Criterion E3.5 (validity check)	`C-B5`	Experiment 3 pre-registration
Measurement-quality criteria P1–P5	`PQ1`–`PQ5`	all three pre-registrations
Figure 5 data	`results.csv` (110 rows)	`experiments/track3_gate2/`
Figure 6 data	`results.csv` (120 rows)	`experiments/track4_gateB/`
Figure 8 data	`results.csv` (800 rows)	`experiments/expB_masking_real/`
Construction module (pinned, Appendix C)	`spurious_cifar.py`	`experiments/expB_masking_real/`
Machine checks (Appendix A)	`verify_xor_min_e.py`	`experiments/track4_gateB/`
Machine checks (Appendix A.4)	`superposition_min_e.py`	`experiments/kef_frontier/`

Table 8 maps every name used in this paper to its archived counterpart. No archived document is needed to follow or verify any claim in this paper.

C Per-Experiment Measurement Details

C.1 Estimator-bias accounting

All tolerances are sized from the asymptotic excess log-loss of a well-specified logistic maximum-likelihood estimator, $\approx d/(2\,n_{\text{train}}\ln 2)$ bits for d fitted parameters. Experiment 1: at most 0.0038 bits (the X arm at $d_n = 256$), plus seed-mean noise $\lesssim 0.001$ bits, giving the committed tolerance $\delta = 0.010$ bits; the worst measured deviations from analytic ground truth (+0.00351, +0.00345 bits at $d_n = 256$, Table 9) match this prediction. Experiment 2 used the same $\delta = 0.010$ bits, with the statistical-power provision of Appendix B.4. Experiment 3's readout has $K_{\text{readout}} = 2570$ parameters fit on $n = 5{,}000$ samples. The asymptotic heuristic $K_{\text{readout}}/(2n\ln 2) \approx 0.3708$ bits is recorded here only as the heuristic that motivated difference-based validation; it is not a validated bias correction: the readout is regularized, the effective degrees of freedom differ from the raw parameter count, and the multinomial parameterization carries gauge redundancy. At this readout dimension the absolute held-out cross-entropy is not a validated estimator of the population optimum: the committed selftest showed absolute recovery failing the required tolerance, while the differences (Δ_{info}, Δ_{transfer}, between-arm gaps) that all pass/fail comparisons consume are recovered to the stated tolerance (Appendix B.5).

C.2 Experiment 1

The committed analytic values (every in-distribution cross-entropy, and the post-shift predictions for the frozen readout) were computed from the generative model by Gauss–Hermite quadrature. The declared in-distribution estimation target is $\mathrm{CE}^{\star}_{\mathcal{A}}$ for the logistic class; for the X and S arms the task is logistic-linear by construction, so the class contains the Bayes readout and $\mathrm{CE}^{\star}_{\mathcal{A}} = H(Y \mid Z)$ exactly; for the N arm the analytic gap between the two is $< 3 \times 10^{-6}$ bits at every d_n. The readout is logistic regression trained by damped Newton (25 iterations, fixed; zero step-halvings occurred, so measured training FLOPs equal the committed formula values). Inference FLOPs: $6.20 \times 10^6 \to 3.00 \times 10^6$ ($d_n = 8$), $2.86 \times 10^7 \to 3.00 \times 10^6$ ($d_n = 64$), $1.054 \times 10^8 \to 3.00 \times 10^6$ ($d_n = 256$) for the $X \to S$ substitution. Measurement-quality outcomes: coefficients of variation across cells were 67% (CE-based uncertainty proxy), 204% (E-FLOPs), 166% (E-wall-clock), 158% (K); all five criteria P1–P5 passed.

C.3 Experiment 2

The per-seed exact floors $\mathrm{CE}^{\star}_{\mathrm{lin}}$ were computed by convex optimization on the exact 1820-atom distribution and committed before the run. The ReLU rung trained with 3 restarts, selection by training loss only; its held-out value estimates the population risk of the trained candidate, which upper-bounds the class optimum $\mathrm{CE}^{\star}_{\mathrm{relu}}$ (no analytic value was committed for this class). Measurement-quality outcomes: coefficients of variation across cells were 134% (CE-based uncertainty proxy), 145% (K), 149% (E_{inf}), 150% (E_{train}), 132% (wall-clock); all five criteria P1–P5 passed. One infrastructure disclosure from the results file is repeated here for completeness: during the run, an asynchronous monitoring channel produced spurious progress notifications inconsistent with the on-disk log; every reported number was read from the committed artifacts after verified process exit, and the artifacts are deterministic given the declared seed streams.

C.4 Experiment 3

Every train, test, and shift image set is built by the pinned construction module `spurious_cifar.py` and no other code; its SHA-256 digest is

`ab8f6e61d54be4ea13b23aee326aabcc`
`b2f272c4bfffa4431ace45b70c9050ba`.

The readout's regularization was pinned at $C = 1.0$ *before the run* for a documented convergence reason: the effectively unregularized maximum-likelihood fit ($C = 10^8$, the harness default) fails to converge on the near-separable CNN features and would have silently corrupted every cross-entropy in the run. Fold scheme: the readout is fit on one stratified half of the test pool (5,000 images) and evaluated held-out on the other half (5,000); the frozen readout is never evaluated on re-tinted versions of images it was fit on; the Q-refit value is a 5-fold cross-fit. The committed tolerance formula was $\delta_{\mathrm{cell}} = \max(2 \times SE_{\mathrm{seed}},\ 0.10 \text{ bits})$, with a statistical-power clause ($\delta_{\mathrm{cell}} > 0.25$ bits degrades any touching criterion to inconclusive; it never fired). Additional checks: the $\rho = 0.9$ coincidence of the two shift distributions agreed at 0.0014 bits (naive arm) and 0.0050 bits (lawful arm); every readout fit converged; the lawful arm's four identical-data replicate trainings per seed put CNN training nondeterminism at ≈ 0.025 bits mean per-seed range, an order of magnitude inside the smallest threshold any criterion uses; cross-seed standard errors of the cell means were 0.0035–0.0588 bits; scikit-learn settings were serialized verbatim into every result row.

D Proofs for Section 3

The statements proved here are classical (Section 6.2); the proofs are relocated from the body unchanged, so that the paper remains self-contained and the a.s. conventions the experiments rely on are checkable.

Proof of Lemma 3.1. Since $Z = \phi(X)$ is $\sigma(X)$-measurable, $\sigma(X, Z) = \sigma(X)$, hence $P(Y = y \mid X, Z) = P(Y = y \mid X) = p(y \mid X)$ a.s.

($\Rightarrow$) Suppose $p(y \mid X) = p(y \mid \phi(X)) =: q(y \mid Z)$ a.s., where $q(y \mid \cdot)$ is $\mathcal{B}_{\mathcal{Z}}$-measurable. Then $q(y \mid Z)$ is a version of $P(Y = y \mid Z)$ by the tower property:

$$\begin{aligned} P(Y = y \mid Z) &= \mathbb{E}\big[P(Y = y \mid X) \,\big|\, Z\big] \\ &= \mathbb{E}\big[q(y \mid Z) \,\big|\, Z\big] = q(y \mid Z) \quad \text{a.s.} \end{aligned}$$

Hence $P(Y = y \mid X, Z) = p(y \mid X) = q(y \mid Z) = P(Y = y \mid Z)$ a.s., which is $Y \perp X \mid Z$.

($\Leftarrow$) If $P(Y = y \mid X, Z) = P(Y = y \mid Z)$ a.s., then $p(y \mid X) = P(Y = y \mid X, Z) = P(Y = y \mid Z) = p(y \mid Z)$ a.s., which is Definition 2.4. ☐

Proof of Lemma 3.2. Three preliminary observations. *(i) The log-arguments are a.s. positive:* $P(p(Y \mid Z) = 0) = \mathbb{E}\big[\sum_y p(y \mid Z)\,\mathbf{1}\{p(y \mid Z) = 0\}\big] = 0$, and likewise $p(Y \mid X) > 0$ a.s. *(ii) Absolute continuity holds a.s.:* by the tower property $p(y \mid Z) = \mathbb{E}[\,p(y \mid X) \mid Z\,]$ a.s.; since $p(y \mid X) \geq 0$, on the event $\{p(y \mid Z) = 0\}$ we have $p(y \mid X) = 0$ a.s. Hence for P_X-a.e. x, $p(\cdot \mid x) \ll p(\cdot \mid \phi(x))$, and the divergence in (1) is a.s. finite. *(iii) Both* *entropies are finite* by (A1), with $\mathbb{E}[-\log p(Y \mid X)] = H(Y \mid X)$ and $\mathbb{E}[-\log p(Y \mid Z)] = H(Y \mid Z)$, so the difference of expectations may be combined into one expectation.

Now compute:

$$
\begin{aligned}
&H(Y \mid Z) - H(Y \mid X) \\
&\qquad = \mathbb{E}\big[-\log p(Y \mid Z)\big] - \mathbb{E}\big[-\log p(Y \mid X)\big] \\
&\qquad = \mathbb{E}\left[\log \frac{p(Y \mid X)}{p(Y \mid Z)}\right].
\end{aligned}
$$

Conditioning on X (under which $Z = \phi(X)$ is fixed) and using the tower property,

$$
\begin{aligned}
&\mathbb{E}\left[\log \frac{p(Y \mid X)}{p(Y \mid Z)} \,\middle|\, X\right] \\
&\qquad = \sum_{y \in \mathcal{Y}} p(y \mid X) \log \frac{p(y \mid X)}{p(y \mid \phi(X))} \\
&\qquad = D\big(p(\cdot \mid X) \,\big\|\, p(\cdot \mid \phi(X))\big) \quad \text{a.s.}
\end{aligned}
$$

Taking expectations gives (1). Nonnegativity of the right side is the information inequality $D \geq 0$ [1, Thm. 2.6.3]; the upper bound follows from (iii). $\square$

Proof of Theorem 3.3. Inequality, always: $Y \to X \to Z$ is a Markov chain by (A2), so $I(Y;Z) \leq I(Y;X)$ is the DPI [1, Thm. 2.8.1]; equivalently, since $H(Y) < \infty$ by (A1), $H(Y \mid Z) = H(Y) - I(Y;Z) \geq H(Y) - I(Y;X) = H(Y \mid X)$. (Lemma 3.2 reproves this: the right side of (1) is nonnegative.)

Equality under sufficiency: suppose $p(y \mid x) = p(y \mid \phi(x))$ for P_X-a.e. x and every y. By Lemma 3.1 ($\Rightarrow$), $p(y \mid \phi(X))$ is a version of $P(Y = y \mid Z)$, so $p(\cdot \mid X) = p(\cdot \mid Z)$ a.s. Then

$$
\begin{aligned}
H(Y \mid X) &= \mathbb{E}\big[\mathcal{H}(p(\cdot \mid X))\big] = \mathbb{E}\big[\mathcal{H}(p(\cdot \mid Z))\big] \\
&= H(Y \mid Z),
\end{aligned}
$$

the middle equality holding because a.s.-equal random variables have equal expectations. (Equivalently: the integrand of (1) is $D(\mu \| \mu) = 0$ a.s.) Finally $I(Y;Z) = H(Y) - H(Y \mid Z) = H(Y) - H(Y \mid X) = I(Y;X)$. $\square$

Proof of Theorem 3.4. By Lemma 3.2, $H(Y \mid Z) - H(Y \mid X) = \mathbb{E}\big[D(p(\cdot \mid X) \,\|\, p(\cdot \mid Z))\big]$. The integrand is nonnegative, and by the equality condition of the information inequality [1, Thm. 2.6.3], $D(\mu \| \nu) = 0$ iff $\mu = \nu$; hence the integrand is strictly positive exactly on $\{X \in A\}$ (up to a null set; it is a.s. finite by Lemma 3.2(ii)). A nonnegative random variable strictly positive on an event of positive probability has strictly positive expectation:

$$
\mathbb{E}\big[D(\cdots)\big] \geq \mathbb{E}\big[D(\cdots)\,\mathbf{1}\{X \in A\}\big] > 0
$$

since $P(X \in A) = P_X(A) > 0$. Therefore $H(Y \mid Z) > H(Y \mid X)$, and with $H(Y) < \infty$, $I(Y;Z) < I(Y;X)$. $\square$

Remark D.1 (stochastic representation maps)*.* Let ϕ instead be a Markov kernel $\kappa(\mathrm{d}z \mid x)$ with Z drawn from $\kappa(\cdot \mid X)$ conditionally independently of Y; (A2) becomes the explicit requirement that $Y \to X \to Z$ is a Markov chain. Sufficiency generalizes to $Y \perp X \mid Z$, and both theorems hold verbatim: the Markov property gives $P(Y = y \mid X, Z) = P(Y = y \mid X)$ a.s., so Lemma 3.2 becomes $H(Y \mid Z) - H(Y \mid X) = I(Y; X \mid Z) = \mathbb{E}[D(p(\cdot \mid X, Z) \,\|\, p(\cdot \mid Z))]$, zero iff $Y \perp X \mid Z$ and strictly positive iff conditional independence fails with positive probability. The deterministic case is recovered by $\kappa(\cdot \mid x) = \delta_{\phi(x)}$.

Proof of Proposition 3.6. Y is unbiased and a deterministic function of Z, so $H(Y) = 1$ and $H(Y \mid Z) = 0$.

Step 1: no $r \in \mathcal{A}_{\text{lin}}$ achieves $H(Y \mid r(Z)) = 0$. A readout $r(z) = \mathbf{1}\{w \cdot z \geq \theta\}$ partitions the four equiprobable points into $B = \{z : w \cdot z \geq \theta\}$ and B^c; zero conditional entropy requires Y a.s. constant on each cell, which (since $H(Y) = 1$) forces B or B^c to equal the fiber $\{(0,1),(1,0)\}$, i.e. linear separability of XOR. Both cases are impossible: if $B = \{(0,1),(1,0)\}$, the defining inequalities are $w_2 \geq \theta$, $w_1 \geq \theta$, $0 < \theta$, $w_1 + w_2 < \theta$, and the first two give $w_1 + w_2 \geq 2\theta > \theta$, a contradiction; if $B = \{(0,0),(1,1)\}$, they are $0 \geq \theta$, $w_1 + w_2 \geq \theta$, $w_1 < \theta$, $w_2 < \theta$, and the last two give $w_1 + w_2 < 2\theta \leq \theta$, a contradiction [6].

Step 2: exact floor by enumeration. The halfplane-achievable partitions of the four points are exactly: trivial ($B = \emptyset$ or everything), singleton/triple, and adjacent pairs (two points differing in one coordinate); the two diagonal pairs are excluded by Step 1, and each remaining subset is realized by an explicit (w, θ) (e.g. $B = \{(1,1)\}$ by $w = (1,1)$, $\theta = 3/2$; $B = \{(1,0),(1,1)\}$ by $w = (1,0)$, $\theta = 1/2$). With all four points of mass 1/4: trivial partitions give $H(Y \mid r(Z)) = H(Y) = 1$; an adjacent pair puts one point of each label in each cell, giving 1; a singleton/triple, e.g. $B = \{(1,1)\}$, gives Y pure on the singleton (probability 1/4) and label proportions 2:1 on the triple (probability 3/4), so $H(Y \mid r(Z)) = \frac{1}{4} \cdot 0 + \frac{3}{4}\, h_2(\frac{1}{3})$, the same value for all eight singleton/triple cells by symmetry. The infimum over $\mathcal{A}_{\text{lin}}$ is the minimum of these finitely many values, $\frac{3}{4} h_2(\frac{1}{3}) > 0$.

Step 3: error probabilities. Trivial and adjacent-pair partitions give best error 1/2; singleton/triple partitions give error 1/4 (predict the pure label on the singleton, the majority label on the triple, erring on one point of mass 1/4). Hence $\min_{r \in \mathcal{A}_{\text{lin}}} P(\hat{Y}(r(Z)) \neq Y) = 1/4 > 0$. □

E Experiment Details: Full Tables and Secondary Observations

This appendix holds the full per-cell tables and pre-declared secondary observations for Experiments 1 and 2, relocated from the body; estimator-bias accounting and measurement-quality outcomes are in Appendix C.

E.1 Experiment 1

Table 9 gives the in-distribution outcomes against the committed analytic values. The analytic sufficient-arm entropy is $H(Y \mid S) = H(Y \mid X) = 0.66654$ bits at every d_n. The measured S-versus-X gaps were +0.00000, +0.00015, +0.00089, +0.00395 bits at $d_n = 0, 8, 64, 256$; the largest matches the predicted estimator bias (0.0038 bits) almost exactly. The counted savings: parameters fell $25 \to 9$ ($\times 2.78$), $137 \to 9$ ($\times 15.2$), $521 \to 9$ ($\times 57.9$); training FLOPs by $\times 4.5$, $\times 101$, $\times 1397$; inference FLOPs from up to 1.054×10^8 to 3.00×10^6.

Table 9: Experiment 1, in-distribution: lawful compression at constant task-relevant H. Measured values are seed-means over all 10 seeds; "dev." is measured minus analytic (committed pre-run). K counts readout parameters plus input dimension; train FLOPs are the deterministic formula values, which the measured counts matched exactly (zero step-halvings occurred).

d_n	arm	K	train FLOPs	$\widehat{\text{CE}}_{\text{ID}}$ (bits)	analytic	dev. (bits)
0	$X{=}S$	9	1.24×10^8	0.66625	0.66654	−0.00029
8	X	25	5.54×10^8	0.66734	0.66654	+0.00080
8	S	9	1.24×10^8	0.66718	0.66654	+0.00065
8	N	17	2.99×10^8	0.67971	0.67885	+0.00086
64	X	137	1.25×10^{10}	0.66697	0.66654	+0.00044
64	S	9	1.24×10^8	0.66608	0.66654	−0.00046
64	N	129	1.12×10^{10}	0.66845	0.66813	+0.00032
256	X	521	1.73×10^{11}	0.67005	0.66654	+0.00351
256	S	9	1.24×10^8	0.66610	0.66654	−0.00044
256	N	513	1.68×10^{11}	0.67039	0.66694	+0.00345

Two pre-declared secondary observations. First, the raw X arm itself drifts under shift (+0.0008, +0.0062, +0.0229 bits at $d_n = 8, 64, 256$): the MLE places small weight on the nuisance direction that is near-collinear with the signal in-distribution, and that direction flips. Discarding the nuisance is thus not only cheaper at equal H; in this toy it removes a fragility (the lawful arm's worst drift is +0.0018 bits). This is an observation about this construction, not a claimed theorem. Second, the committed failure conditions (an insufficient arm beating the sufficient arm, in-distribution or under shift) did not occur at any cell (closest approach: the N arm's cross-entropy exceeded the S arm's everywhere). The committed checks on measurement quality all passed (Appendix C).

E.2 Experiment 2

Table 10 gives the measured rungs against the committed exact floors; Table 11 the declared ladder accounting. Construction constants: $Y = F_1$ with $H(Y) = h_2(0.25) = 0.8113$ bits; the per-seed projection $A \in \mathbb{R}^{d_z \times 16}$ has i.i.d. standard normal entries and unit-norm columns; minimum pairwise atom distance 0.0416 over all 40 cells (the injectivity check). The linear rung's paired standard errors against its per-seed floors were 0.0002–0.0007 bits; the worst single-seed excursion below a floor anywhere was 0.0032 bits, inside the 0.010-bit tolerance, and no cheaper rung beat a costlier one by more than the tolerance.

Table 10: Experiment 2: measured held-out cross-entropy (bits, seed-means over 10 seeds) against the committed exact $\mathrm{CE}^{\star}_{\mathrm{lin}}$ floors, and the declared counted accounting. floor* is the seed-mean of the per-seed exact population optimum of the linear-logistic class ($\mathrm{CE}^{\star}_{\mathrm{lin}}$, Definition 2.3). Rung (iii)'s 1.44×10^{-6} bits is exactly the pre-registered probability-clipping artifact ($-\log_2(1 - 10^{-6})$ at clip 10^{-6}).

d_z	load	floor*	(i) linear	(ii) ReLU-64	(iii) oracle
4	4.00	0.6053	0.6066	0.3729	1.44×10^{-6}
6	2.67	0.4799	0.4798	0.0569	1.44×10^{-6}
8	2.00	0.3196	0.3196	0.0007	1.44×10^{-6}
12	1.33	0.0666	0.0668	3.6×10^{-6}	1.44×10^{-6}

Table 11: Experiment 2: declared ladder accounting (fixed before the run; K in stored parameters / codebook entries, all unshared counts K_{unsh}, no parameter sharing in any rung; and E_{inf} in FLOPs per sample). K and E_{inf} strictly increase up-ladder at every cell by construction.

d_z	K_{i}	K_{ii}	K_{iii}	E_{i}	E_{ii}	E_{iii}
4	5	385	9100	15	839	23661
6	7	513	12740	19	1095	34581
8	9	641	16380	23	1351	45501
12	13	897	23660	31	1863	67341